house-price-prediction

March 28, 2026

[2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[4]:
```
df = pd.read_csv(r"F:\\data sets\\house_price - house_price.csv")
df.head()
```

[4]:
```
   Order        PID  MS SubClass MS Zoning  Lot Frontage  Lot Area Street  \
0      1  526301100           20        RL         141.0     31770   Pave
1      2  526350040           20        RH          80.0     11622   Pave
2      3  526351010           20        RL          81.0     14267   Pave
3      4  526353030           20        RL          93.0     11160   Pave
4      5  527105010           60        RL          74.0     13830   Pave

  Alley Lot Shape Land Contour  … Pool Area Pool QC  Fence Misc Feature  \
0   NaN       IR1          Lvl  …         0     NaN    NaN          NaN
1   NaN       Reg          Lvl  …         0     NaN  MnPrv          NaN
2   NaN       IR1          Lvl  …         0     NaN    NaN         Gar2
3   NaN       Reg          Lvl  …         0     NaN    NaN          NaN
4   NaN       IR1          Lvl  …         0     NaN  MnPrv          NaN

  Misc Val Mo Sold Yr Sold Sale Type  Sale Condition  SalePrice
0        0       5    2010        WD          Normal     215000
1        0       6    2010        WD          Normal     105000
2    12500       6    2010        WD          Normal     172000
3        0       4    2010        WD          Normal     244000
4        0       3    2010        WD          Normal     189900

[5 rows x 82 columns]
```

[6]:
```
df.shape
```

[6]:
```
(2930, 82)
```

[8]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2930 entries, 0 to 2929
```

```
Data columns (total 82 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Order            2930 non-null   int64
 1   PID              2930 non-null   int64
 2   MS SubClass      2930 non-null   int64
 3   MS Zoning        2930 non-null   str
 4   Lot Frontage     2440 non-null   float64
 5   Lot Area         2930 non-null   int64
 6   Street           2930 non-null   str
 7   Alley            198 non-null    str
 8   Lot Shape        2930 non-null   str
 9   Land Contour     2930 non-null   str
 10  Utilities        2930 non-null   str
 11  Lot Config       2930 non-null   str
 12  Land Slope       2930 non-null   str
 13  Neighborhood     2930 non-null   str
 14  Condition 1      2930 non-null   str
 15  Condition 2      2930 non-null   str
 16  Bldg Type        2930 non-null   str
 17  House Style      2930 non-null   str
 18  Overall Qual     2930 non-null   int64
 19  Overall Cond     2930 non-null   int64
 20  Year Built       2930 non-null   int64
 21  Year Remod/Add   2930 non-null   int64
 22  Roof Style       2930 non-null   str
 23  Roof Matl        2930 non-null   str
 24  Exterior 1st     2930 non-null   str
 25  Exterior 2nd     2930 non-null   str
 26  Mas Vnr Type     1155 non-null   str
 27  Mas Vnr Area     2907 non-null   float64
 28  Exter Qual       2930 non-null   str
 29  Exter Cond       2930 non-null   str
 30  Foundation       2930 non-null   str
 31  Bsmt Qual        2850 non-null   str
 32  Bsmt Cond        2850 non-null   str
 33  Bsmt Exposure    2847 non-null   str
 34  BsmtFin Type 1   2850 non-null   str
 35  BsmtFin SF 1     2929 non-null   float64
 36  BsmtFin Type 2   2849 non-null   str
 37  BsmtFin SF 2     2929 non-null   float64
 38  Bsmt Unf SF      2929 non-null   float64
 39  Total Bsmt SF    2929 non-null   float64
 40  Heating          2930 non-null   str
 41  Heating QC       2930 non-null   str
 42  Central Air      2930 non-null   str
 43  Electrical       2929 non-null   str
 44  1st Flr SF       2930 non-null   int64
```

```
 45  2nd Flr SF       2930 non-null   int64
 46  Low Qual Fin SF  2930 non-null   int64
 47  Gr Liv Area      2930 non-null   int64
 48  Bsmt Full Bath   2928 non-null   float64
 49  Bsmt Half Bath   2928 non-null   float64
 50  Full Bath        2930 non-null   int64
 51  Half Bath        2930 non-null   int64
 52  Bedroom AbvGr    2930 non-null   int64
 53  Kitchen AbvGr    2930 non-null   int64
 54  Kitchen Qual     2930 non-null   str
 55  TotRms AbvGrd    2930 non-null   int64
 56  Functional       2930 non-null   str
 57  Fireplaces       2930 non-null   int64
 58  Fireplace Qu     1508 non-null   str
 59  Garage Type      2773 non-null   str
 60  Garage Yr Blt    2771 non-null   float64
 61  Garage Finish    2771 non-null   str
 62  Garage Cars      2929 non-null   float64
 63  Garage Area      2929 non-null   float64
 64  Garage Qual      2771 non-null   str
 65  Garage Cond      2771 non-null   str
 66  Paved Drive      2930 non-null   str
 67  Wood Deck SF     2930 non-null   int64
 68  Open Porch SF    2930 non-null   int64
 69  Enclosed Porch   2930 non-null   int64
 70  3Ssn Porch       2930 non-null   int64
 71  Screen Porch     2930 non-null   int64
 72  Pool Area        2930 non-null   int64
 73  Pool QC          13 non-null     str
 74  Fence            572 non-null    str
 75  Misc Feature     106 non-null    str
 76  Misc Val         2930 non-null   int64
 77  Mo Sold          2930 non-null   int64
 78  Yr Sold          2930 non-null   int64
 79  Sale Type        2930 non-null   str
 80  Sale Condition   2930 non-null   str
 81  SalePrice        2930 non-null   int64
dtypes: float64(11), int64(28), str(43)
memory usage: 2.2 MB
```

1 Data Cleaning

checking the null values

[12]:
```
columns = list(df.columns)
null_clo = []
for i in columns:
    if df[i].isnull().sum() != 0:
```

```
        print(i,'->',df[i].isnull().sum())
        null_clo.append(i)
```

```
Lot Frontage -> 490
Alley -> 2732
Mas Vnr Type -> 1775
Mas Vnr Area -> 23
Bsmt Qual -> 80
Bsmt Cond -> 80
Bsmt Exposure -> 83
BsmtFin Type 1 -> 80
BsmtFin SF 1 -> 1
BsmtFin Type 2 -> 81
BsmtFin SF 2 -> 1
Bsmt Unf SF -> 1
Total Bsmt SF -> 1
Electrical -> 1
Bsmt Full Bath -> 2
Bsmt Half Bath -> 2
Fireplace Qu -> 1422
Garage Type -> 157
Garage Yr Blt -> 159
Garage Finish -> 159
Garage Cars -> 1
Garage Area -> 1
Garage Qual -> 159
Garage Cond -> 159
Pool QC -> 2917
Fence -> 2358
Misc Feature -> 2824
```

Pool QC -> 2917

Fence -> 2358

Misc Feature -> 2824

These column nearly has 90% null values so i am droping these columnns

[14]:
```
df.drop(columns = ['Pool QC','Fence','Misc Feature'],inplace=True)
df.shape
```

[14]: (2930, 79)

[30]:
```
df.drop(columns=['Fireplace Qu','Mas Vnr Type','Alley'], inplace=True)
```

BsmtFin SF 1 -> 1 BsmtFin SF 2 -> 1 Bsmt Unf SF -> 1 Total Bsmt SF -> 1 Electrical -> 1 Bsmt Full Bath -> 2 Bsmt Half Bath -> 2 Garage Cars -> 1 Garage Area -> 1

[14]:
```
# Removed null values with ffill() method
col=['BsmtFin SF 1','BsmtFin SF 2','Bsmt Unf SF','Total Bsmt␣
↪SF','Electrical','Bsmt Full Bath','Bsmt Half Bath','Garage Cars',
'Garage Area' ]
for i in col:
    df[i]=df[i].ffill()
    print(i,'->',df[i].isnull().sum())
```

```
BsmtFin SF 1 -> 0
BsmtFin SF 2 -> 0
Bsmt Unf SF -> 0
Total Bsmt SF -> 0
Electrical -> 0
Bsmt Full Bath -> 0
Bsmt Half Bath -> 0
Garage Cars -> 0
Garage Area -> 0
```

Replaced integer column null values with the mean of respective column

[16]:
```
# Replaced integer column null values with the mean of respective column
int_col = list(df.select_dtypes(include='int').columns)
#int_col
for i in int_col:
    df[i]=df[i].fillna(df[i].mean())
    print(i,'->',df[i].isnull().sum())
```

```
Order -> 0
PID -> 0
MS SubClass -> 0
Lot Area -> 0
Overall Qual -> 0
Overall Cond -> 0
Year Built -> 0
Year Remod/Add -> 0
1st Flr SF -> 0
2nd Flr SF -> 0
Low Qual Fin SF -> 0
Gr Liv Area -> 0
Full Bath -> 0
Half Bath -> 0
Bedroom AbvGr -> 0
Kitchen AbvGr -> 0
TotRms AbvGrd -> 0
Fireplaces -> 0
Wood Deck SF -> 0
Open Porch SF -> 0
Enclosed Porch -> 0
```

```
3Ssn Porch -> 0
Screen Porch -> 0
Pool Area -> 0
Misc Val -> 0
Mo Sold -> 0
Yr Sold -> 0
SalePrice -> 0
```

Replaced float column null values with the mean of respective column

[19]:
```
float_col = list(df.select_dtypes(include='float').columns)
#float_col
for i in float_col:
    df[i]=df[i].fillna(df[i].mean())
    print(i,'->',df[i].isnull().sum())
```

```
Lot Frontage -> 0
Mas Vnr Area -> 0
BsmtFin SF 1 -> 0
BsmtFin SF 2 -> 0
Bsmt Unf SF -> 0
Total Bsmt SF -> 0
Bsmt Full Bath -> 0
Bsmt Half Bath -> 0
Garage Yr Blt -> 0
Garage Cars -> 0
Garage Area -> 0
```

[21]:
```
columns = list(df.columns)
remaning_null = []
for i in columns:
    if df[i].isnull().sum() != 0:
        print(i,'->',df[i].isnull().sum())
        remaning_null.append(i)
```

```
Alley -> 2732
Mas Vnr Type -> 1775
Bsmt Qual -> 80
Bsmt Cond -> 80
Bsmt Exposure -> 83
BsmtFin Type 1 -> 80
BsmtFin Type 2 -> 81
Fireplace Qu -> 1422
Garage Type -> 157
Garage Finish -> 159
Garage Qual -> 159
Garage Cond -> 159
Pool QC -> 2917
```

```
Fence -> 2358
Misc Feature -> 2824
```

Replaced object datatype column null values with the mode of respective column

[24]:
```
for i in remaning_null:
    df[i]=df[i].fillna(df[i].mode()[0])
    print(i,'->',df[i].isnull().sum())
```

```
Alley -> 0
Mas Vnr Type -> 0
Bsmt Qual -> 0
Bsmt Cond -> 0
Bsmt Exposure -> 0
BsmtFin Type 1 -> 0
BsmtFin Type 2 -> 0
Fireplace Qu -> 0
Garage Type -> 0
Garage Finish -> 0
Garage Qual -> 0
Garage Cond -> 0
Pool QC -> 0
Fence -> 0
Misc Feature -> 0
```

[26]:
```
df.isnull().sum()
```

[26]:
```
Order             0
PID               0
MS SubClass       0
MS Zoning         0
Lot Frontage      0
                 ..
Mo Sold           0
Yr Sold           0
Sale Type         0
Sale Condition    0
SalePrice         0
Length: 82, dtype: int64
```

[32]:
```
df.shape
```

[32]:
```
(2930, 79)
```

2 Handling Duplicate data

[35]:
```
df.duplicated().sum()
```

[35]: 0

There are no duplicate values to handel

3 Cleaned Dataset

[39]:
```
df
```

[39]:
```
      Order        PID  MS SubClass MS Zoning  Lot Frontage  Lot Area Street  \
0         1  526301100           20        RL     141.00000     31770   Pave
1         2  526350040           20        RH      80.00000     11622   Pave
2         3  526351010           20        RL      81.00000     14267   Pave
3         4  526353030           20        RL      93.00000     11160   Pave
4         5  527105010           60        RL      74.00000     13830   Pave
…      …         …           …        …            …        …     …
2925   2926  923275080           80        RL      37.00000      7937   Pave
2926   2927  923276100           20        RL      69.22459      8885   Pave
2927   2928  923400125           85        RL      62.00000     10441   Pave
2928   2929  924100070           20        RL      77.00000     10010   Pave
2929   2930  924151050           60        RL      74.00000      9627   Pave

     Lot Shape Land Contour Utilities  … Pool Area Pool QC  Fence  \
0          IR1          Lvl    AllPub  …         0      Ex  MnPrv
1          Reg          Lvl    AllPub  …         0      Ex  MnPrv
2          IR1          Lvl    AllPub  …         0      Ex  MnPrv
3          Reg          Lvl    AllPub  …         0      Ex  MnPrv
4          IR1          Lvl    AllPub  …         0      Ex  MnPrv
…        …           …       …  …       …      …    …
2925       IR1          Lvl    AllPub  …         0      Ex  GdPrv
2926       IR1          Low    AllPub  …         0      Ex  MnPrv
2927       Reg          Lvl    AllPub  …         0      Ex  MnPrv
2928       Reg          Lvl    AllPub  …         0      Ex  MnPrv
2929       Reg          Lvl    AllPub  …         0      Ex  MnPrv

     Misc Feature Misc Val Mo Sold Yr Sold  Sale Type  Sale Condition  \
0            Shed        0       5    2010         WD          Normal
1            Shed        0       6    2010         WD          Normal
2            Gar2    12500       6    2010         WD          Normal
3            Shed        0       4    2010         WD          Normal
4            Shed        0       3    2010         WD          Normal
…           …      …      …     …        …             …
2925         Shed        0       3    2006         WD          Normal
```

```
2926         Shed        0       6    2006         WD          Normal
2927         Shed      700       7    2006         WD          Normal
2928         Shed        0       4    2006         WD          Normal
2929         Shed        0      11    2006         WD          Normal

      SalePrice
0        215000
1        105000
2        172000
3        244000
4        189900
…         …
2925     142500
2926     131000
2927     132000
2928     170000
2929     188000

[2930 rows x 79 columns]
```

[41]:
```
df.describe().round()
```

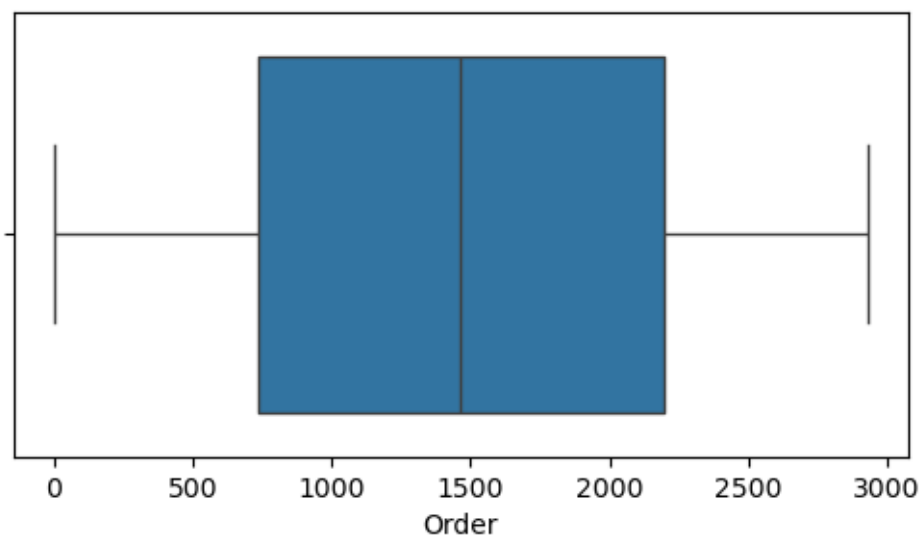
[41]:
```
         Order           PID  MS SubClass  Lot Frontage  Lot Area  \
count  2930.0  2.930000e+03       2930.0        2930.0    2930.0
mean   1466.0  7.144645e+08         57.0          69.0   10148.0
std     846.0  1.887308e+08         43.0          21.0    7880.0
min       1.0  5.263011e+08         20.0          21.0    1300.0
25%     733.0  5.284770e+08         20.0          60.0    7440.0
50%    1466.0  5.354536e+08         50.0          69.0    9436.0
75%    2198.0  9.071811e+08         70.0          78.0   11555.0
max    2930.0  1.007100e+09        190.0         313.0  215245.0

       Overall Qual  Overall Cond  Year Built  Year Remod/Add  Mas Vnr Area  \
count        2930.0        2930.0      2930.0          2930.0        2930.0
mean            6.0           6.0      1971.0          1984.0         102.0
std             1.0           1.0        30.0            21.0         178.0
min             1.0           1.0      1872.0          1950.0           0.0
25%             5.0           5.0      1954.0          1965.0           0.0
50%             6.0           5.0      1973.0          1993.0           0.0
75%             7.0           6.0      2001.0          2004.0         163.0
max            10.0           9.0      2010.0          2010.0        1600.0

       …  Wood Deck SF  Open Porch SF  Enclosed Porch  3Ssn Porch  \
count  …        2930.0         2930.0          2930.0      2930.0
mean   …          94.0           48.0            23.0         3.0
std    …         126.0           67.0            64.0        25.0
min    …           0.0            0.0             0.0         0.0
```

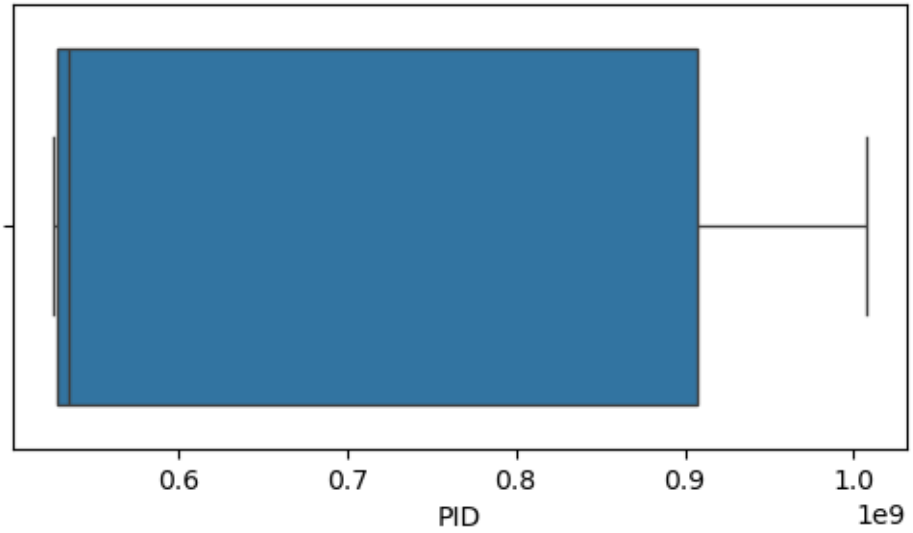
```
25%    …          0.0          0.0          0.0        0.0
50%    …          0.0         27.0          0.0        0.0
75%    …        168.0         70.0          0.0        0.0
max    …       1424.0        742.0       1012.0      508.0

       Screen Porch  Pool Area  Misc Val  Mo Sold  Yr Sold  SalePrice
count        2930.0     2930.0    2930.0   2930.0   2930.0     2930.0
mean           16.0        2.0      51.0      6.0   2008.0   180796.0
std            56.0       36.0     566.0      3.0      1.0    79887.0
min             0.0        0.0       0.0      1.0   2006.0    12789.0
25%             0.0        0.0       0.0      4.0   2007.0   129500.0
50%             0.0        0.0       0.0      6.0   2008.0   160000.0
75%             0.0        0.0       0.0      8.0   2009.0   213500.0
max           576.0      800.0   17000.0     12.0   2010.0   755000.0

[8 rows x 39 columns]
```

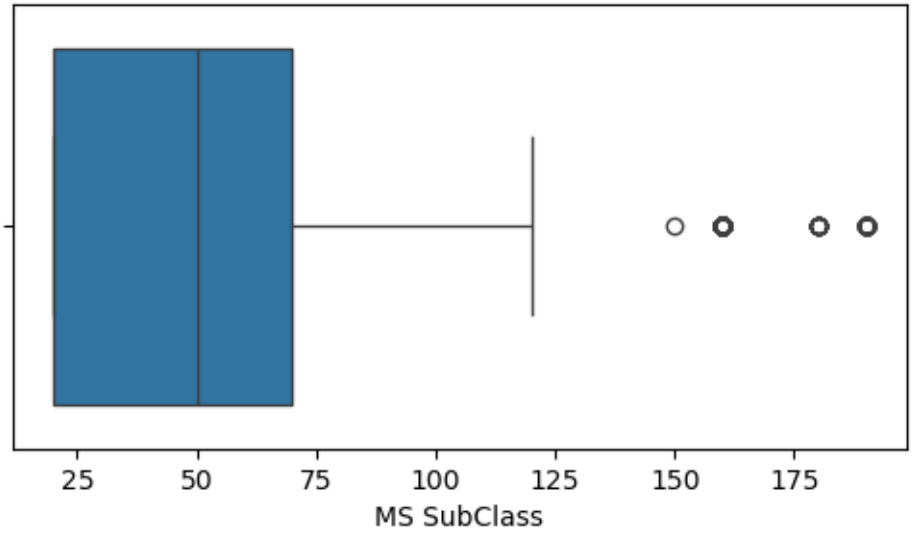
4 Outlier detection

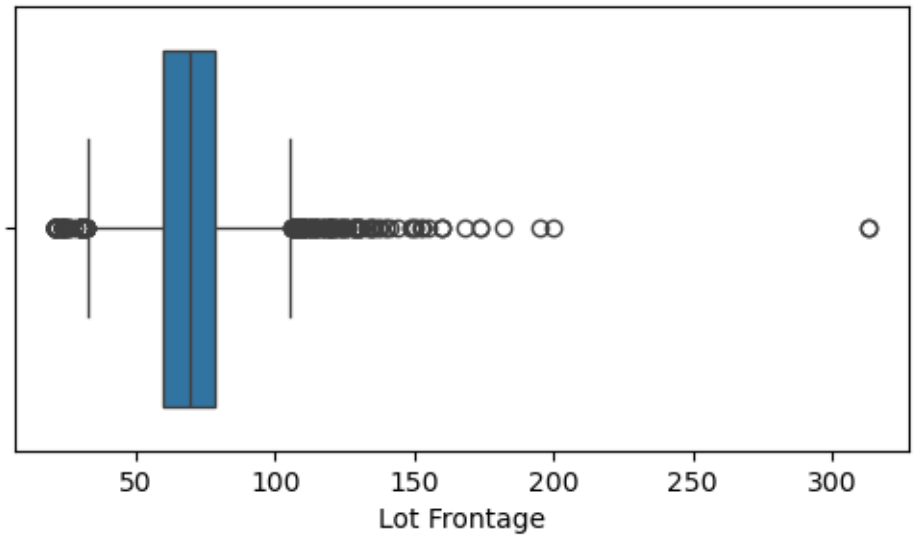
[44]:
```
num_cols = df.select_dtypes(include='number').columns
```

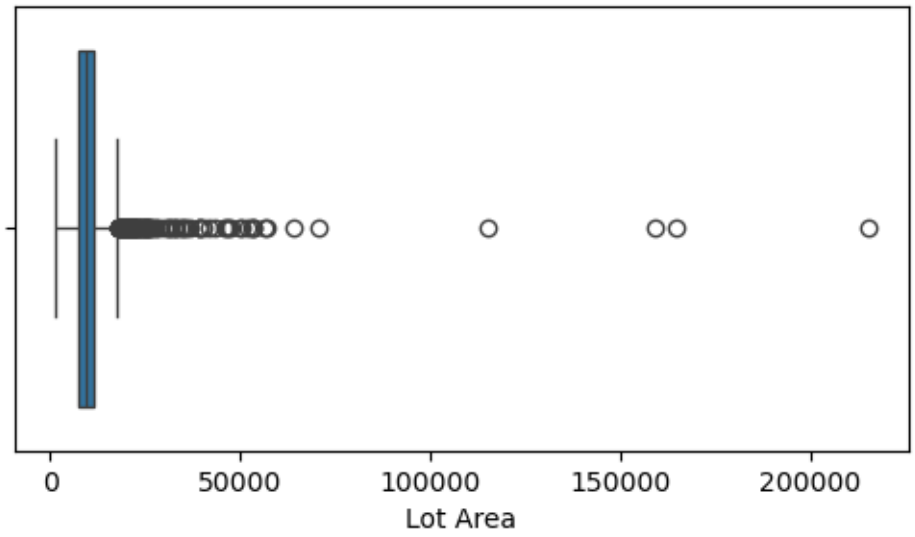
[53]:
```
for i in num_cols:
    plt.figure(figsize=(6,3))
    sns.boxplot(x=df[i])
    plt.show()
```

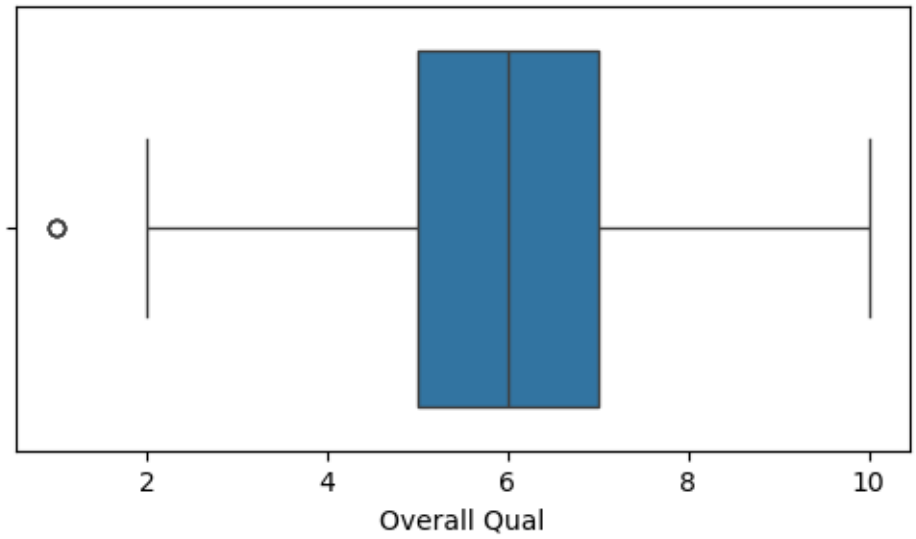


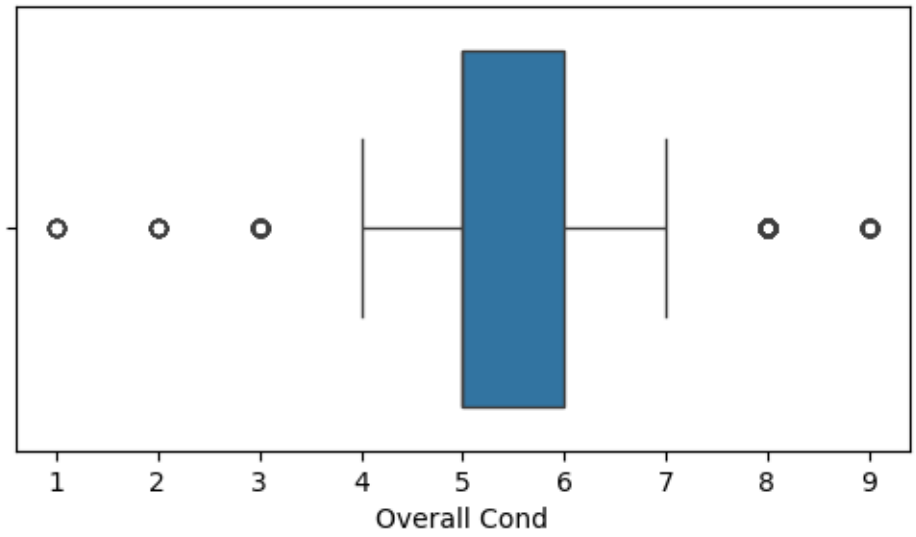

0.6
0.7
0.8
0.9
1.0
PID
1e9

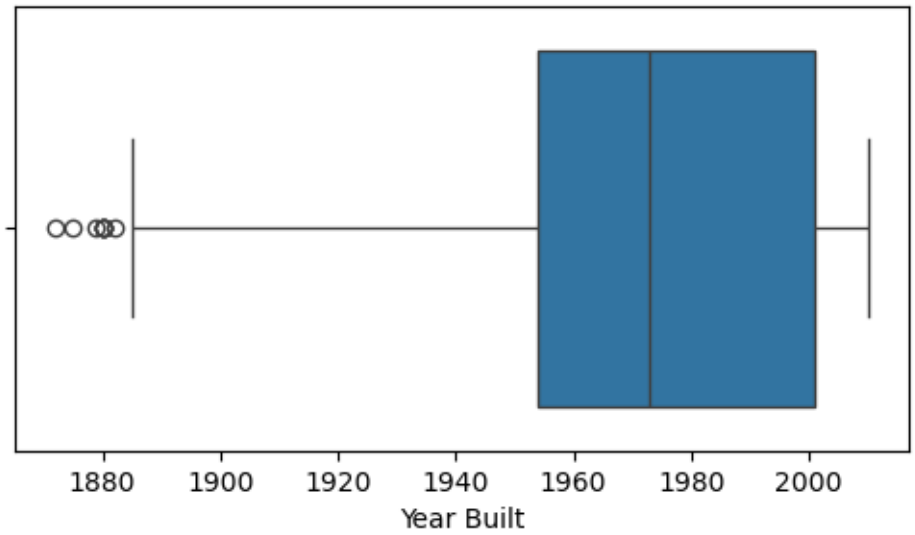

25
50
75
100
125
150
175
MS SubClass

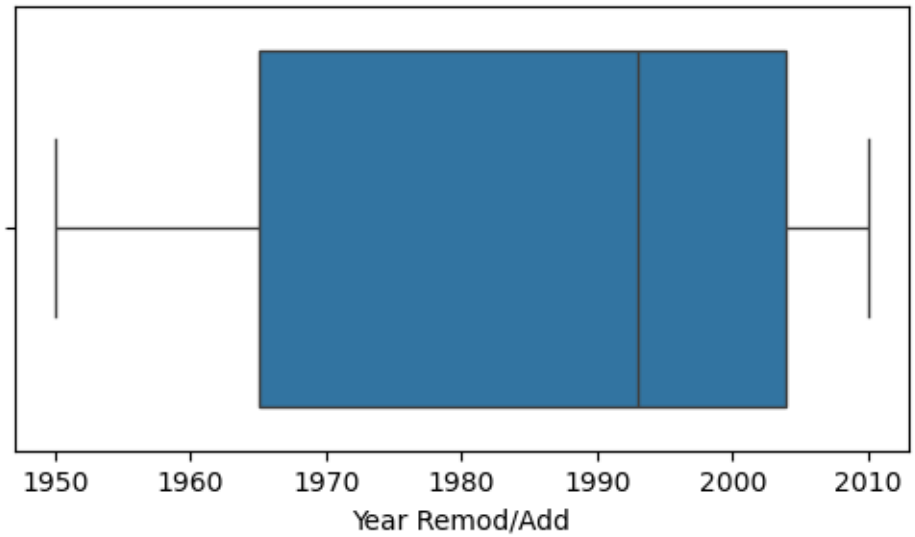

50
100
150
200
250
300
Lot Frontage

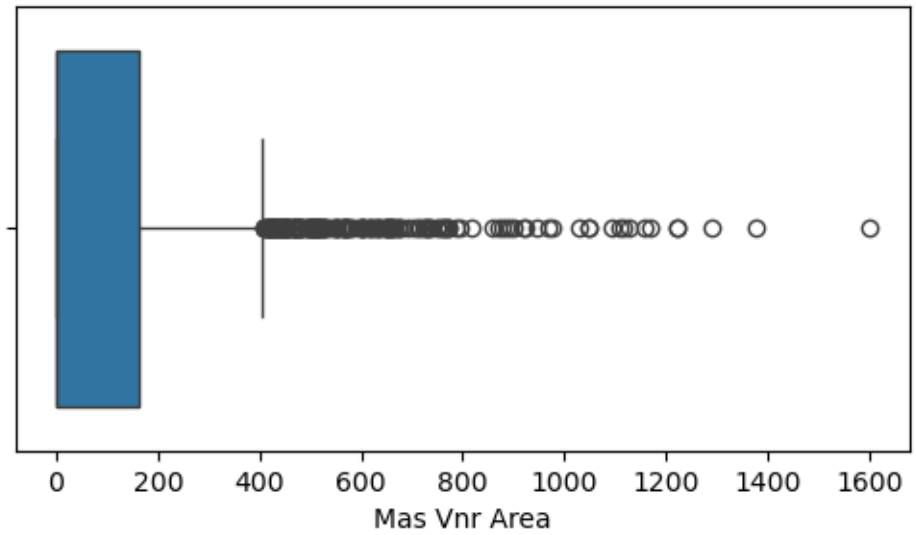

0
50000
100000
150000
200000
Lot Area

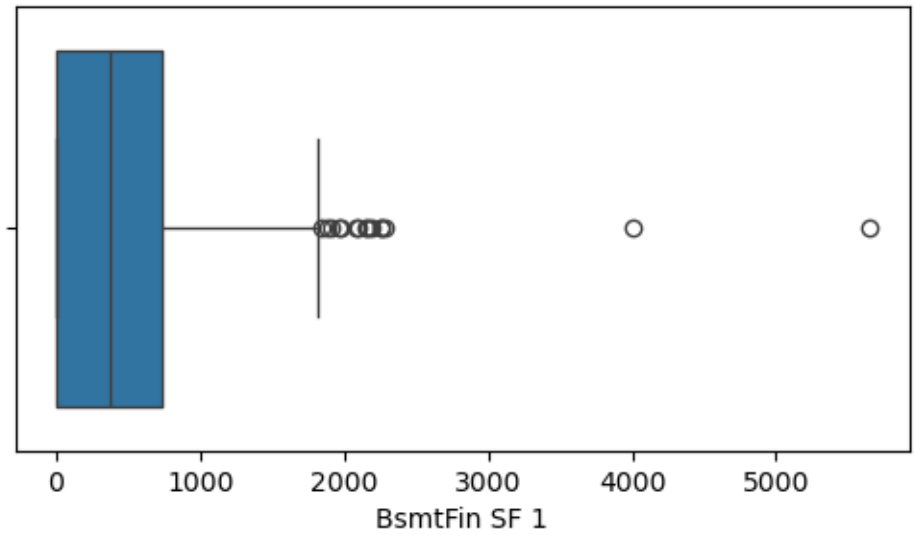

2
4
6
8
10
Overall Qual

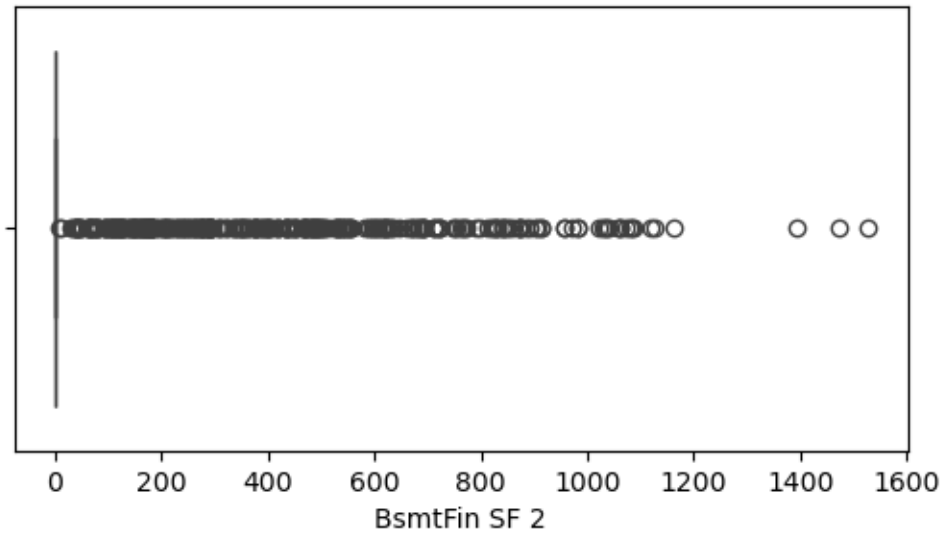

1
2
3
4
5
6
7
8
9
Overall Cond

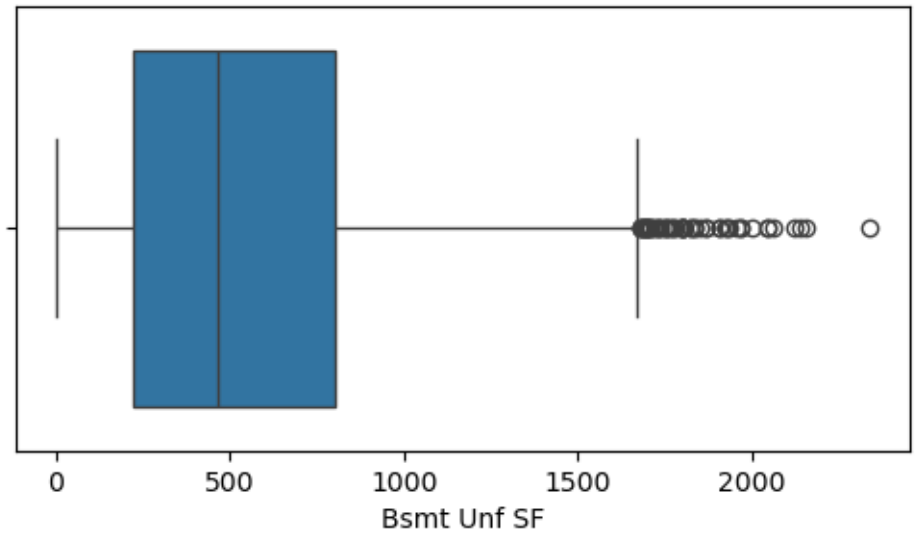

1880
1900
1920
1940
1960
1980
2000
Year Built

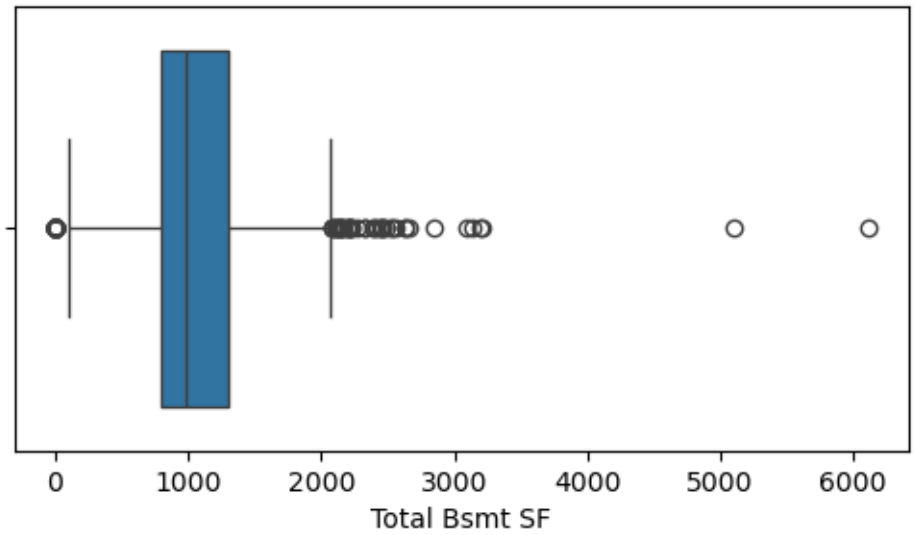

1950
1960
1970
1980
1990
2000
2010
Year Remod/Add

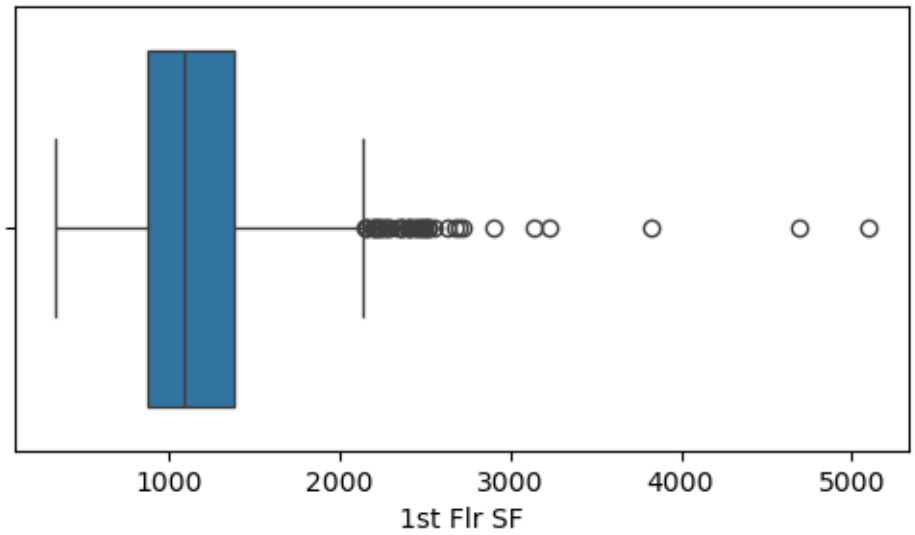

0
200
400
600
800
1000
1200
1400
1600
Mas Vnr Area

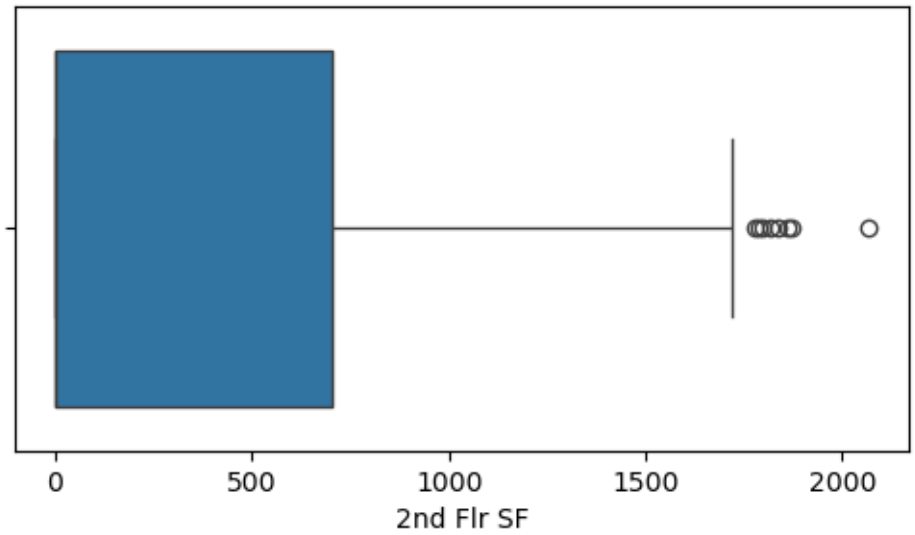

0
1000
2000
3000
4000
5000
BsmtFin SF 1


0
200
400
600
800
1000
1200
1400
1600
BsmtFin SF 2


0
500
1000
1500
2000
Bsmt Unf SF


0
1000
2000
3000
4000
5000
6000
Total Bsmt SF


1000
2000
3000
4000
5000
1st Flr SF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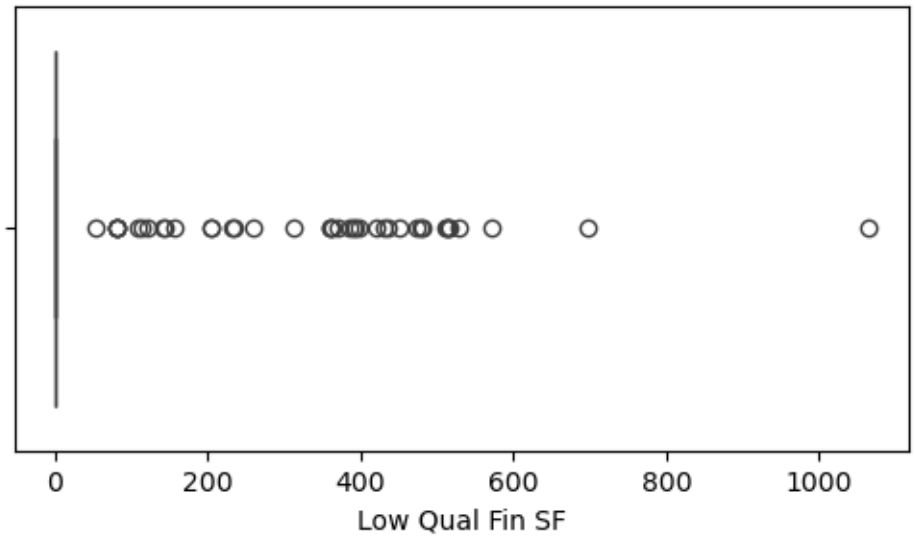
500
1000
1500
2000
2nd Flr SF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200
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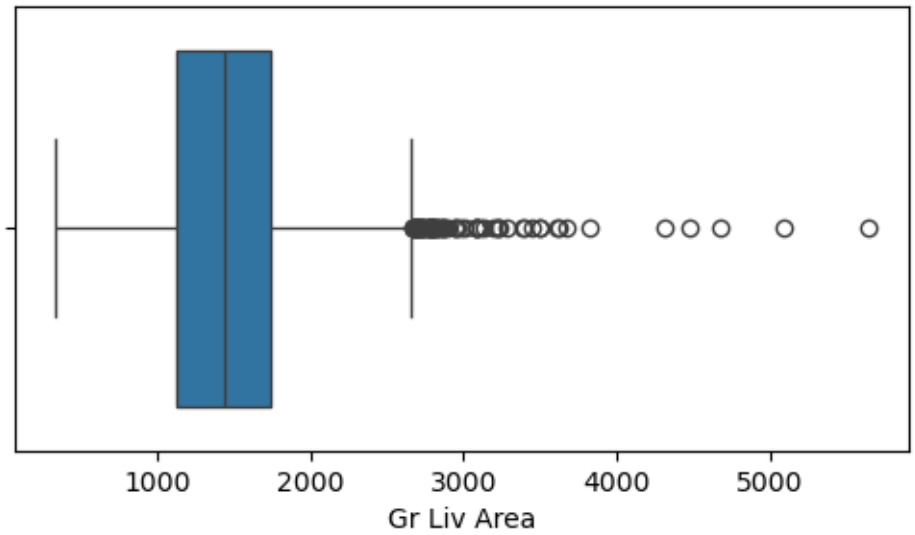
800
1000
Low Qual Fin SF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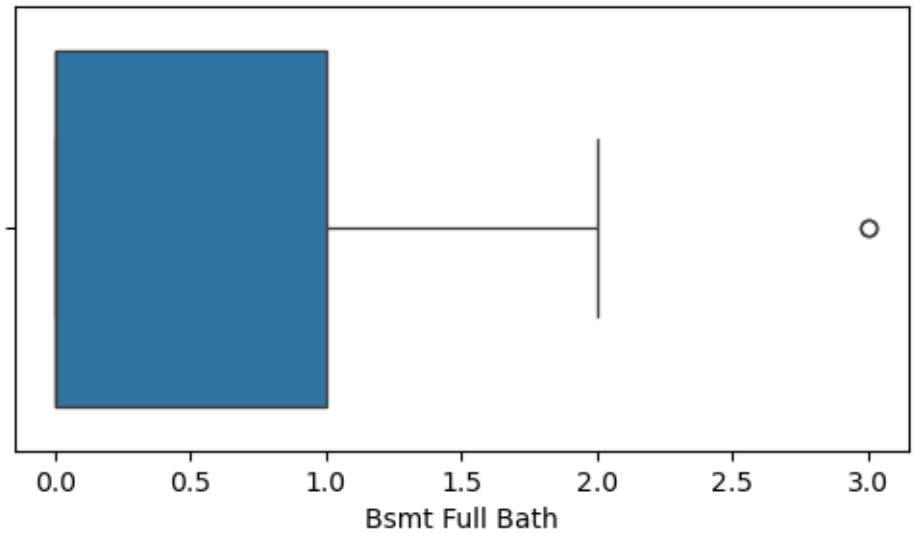
2000
3000
4000
5000
Gr Liv Area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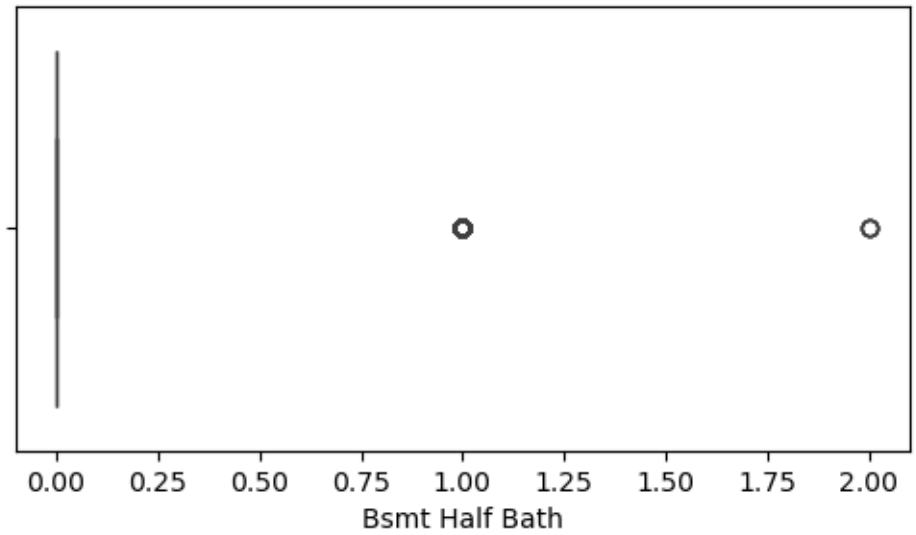
1.0
1.5
2.0
2.5
3.0
Bsmt Full Bath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0.25
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1.00
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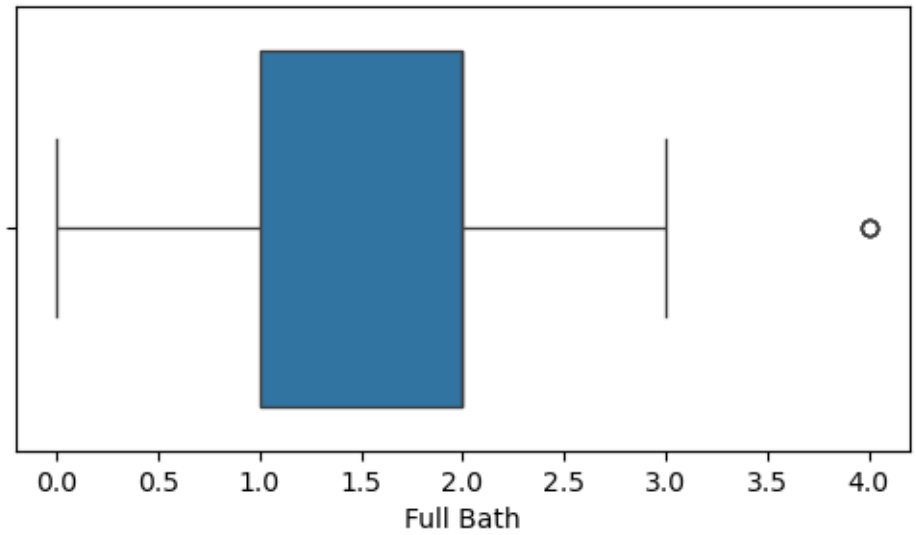
1.50
1.75
2.00
Bsmt Half Bath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4.0
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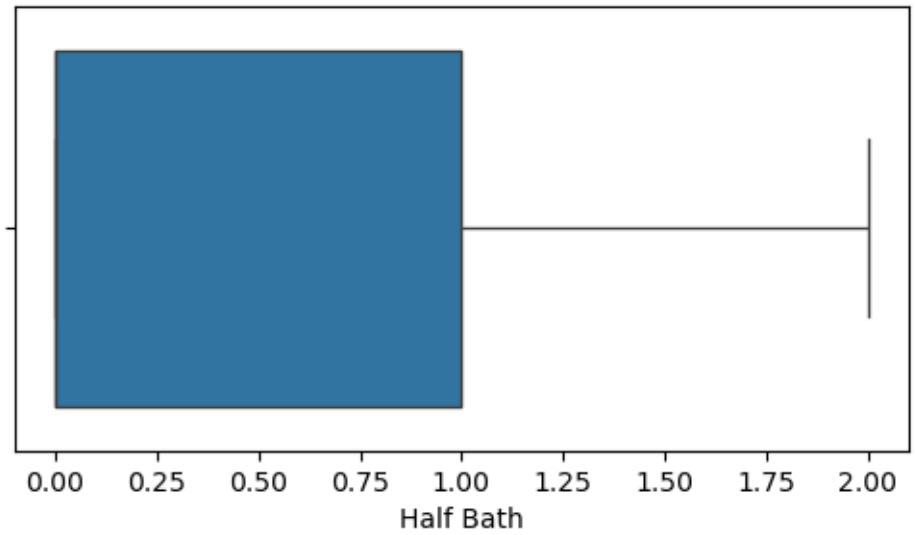
0.00
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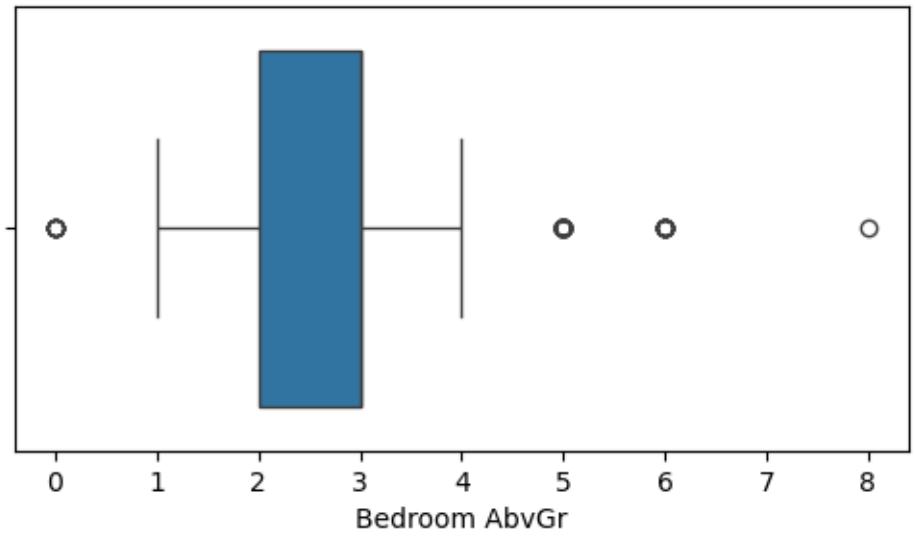
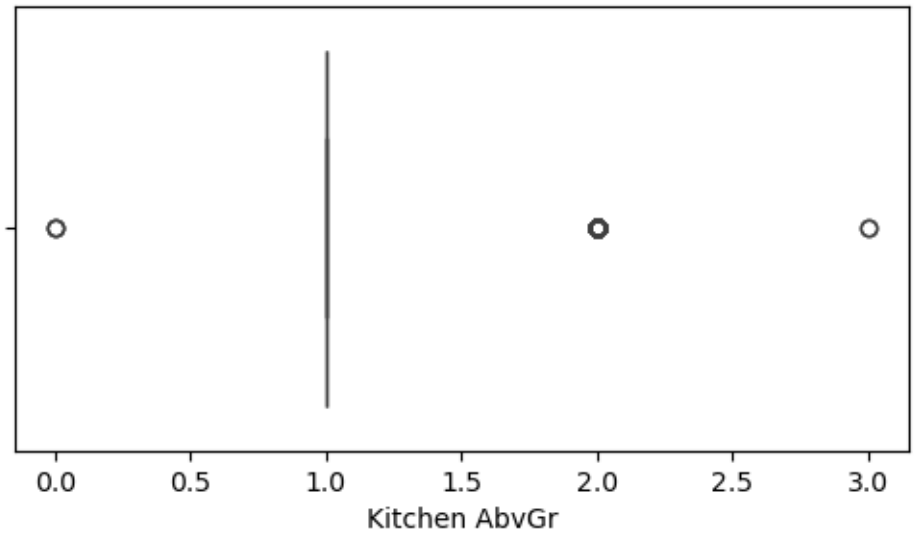
0.75
1.00
1.25
1.50
1.75
2.00
Half Bath


0
1
2
3
4
5
6
7
8
Bedroom AbvGr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0.5
1.0
1.5
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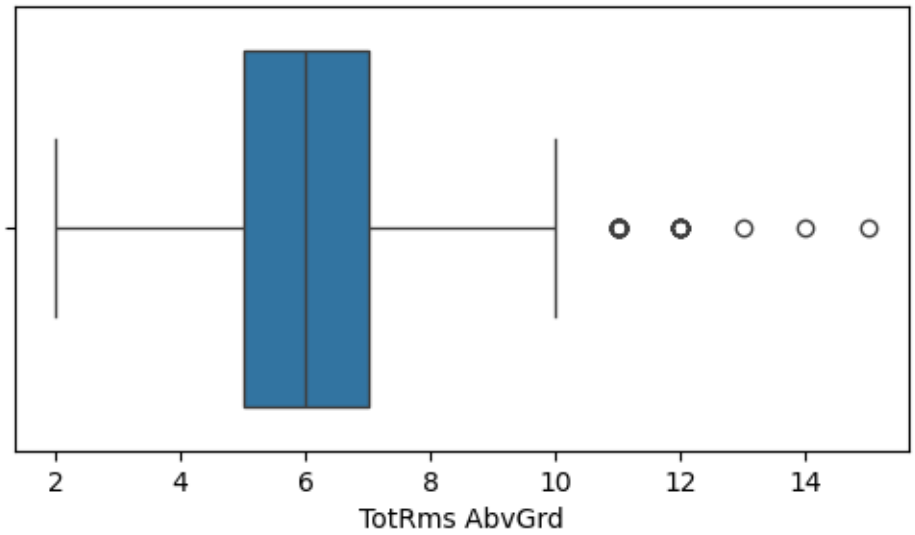
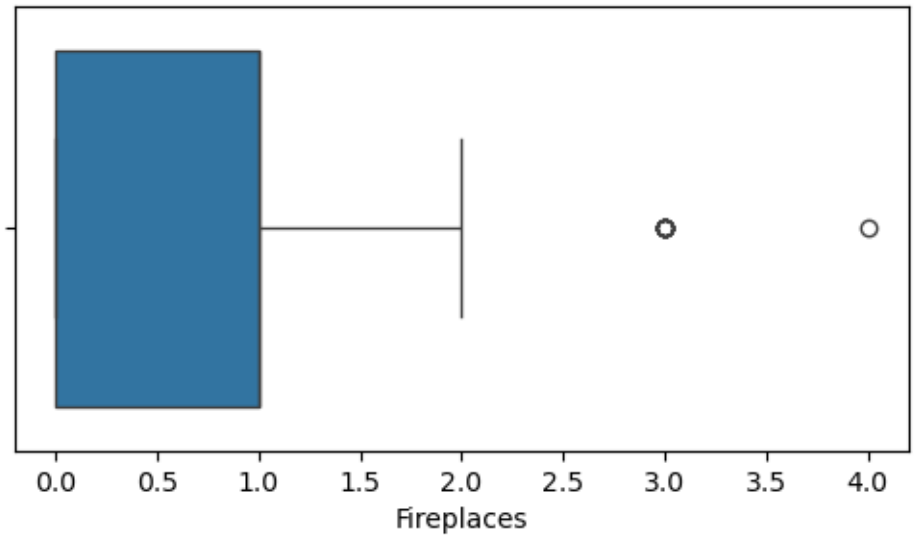
2.5
3.0
Kitchen AbvGr


2
4
6
8
10
12
14
TotRms AbvGrd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0.5
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2.5
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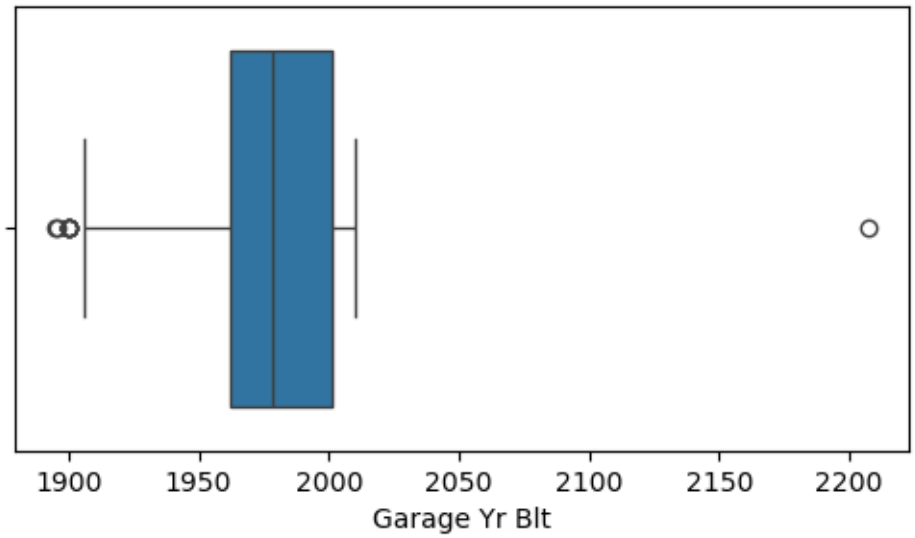
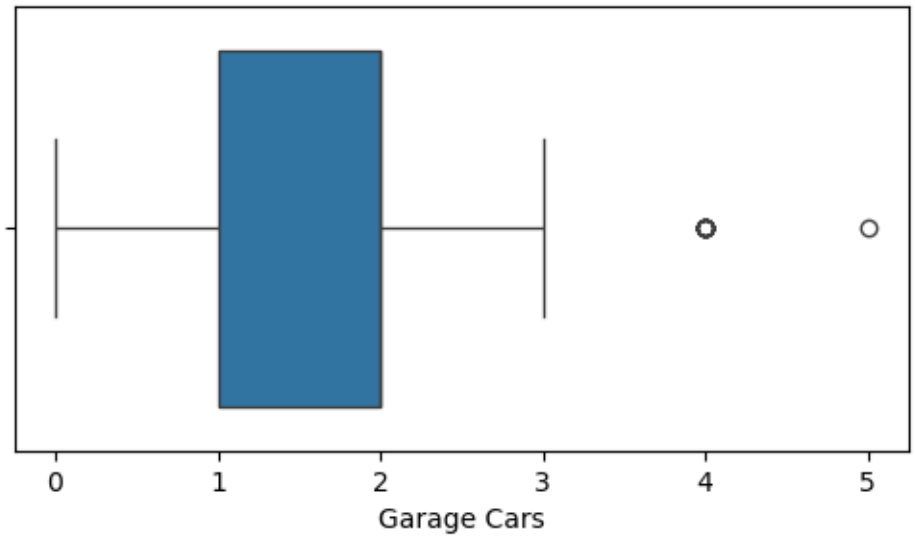
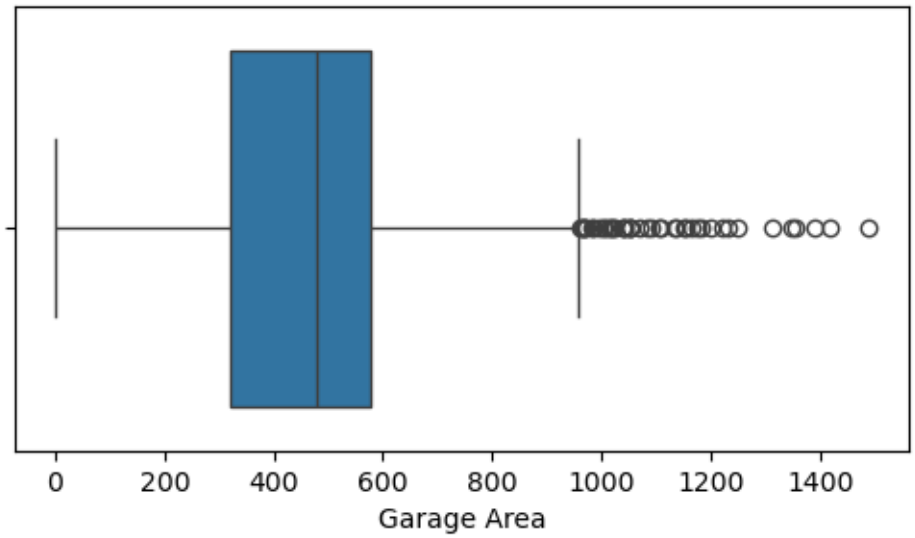
3.5
4.0
Fireplaces


1900
1950
2000
2050
2100
2150
2200
Garage Yr Blt


0
1
2
3
4
5
Garage Cars


0
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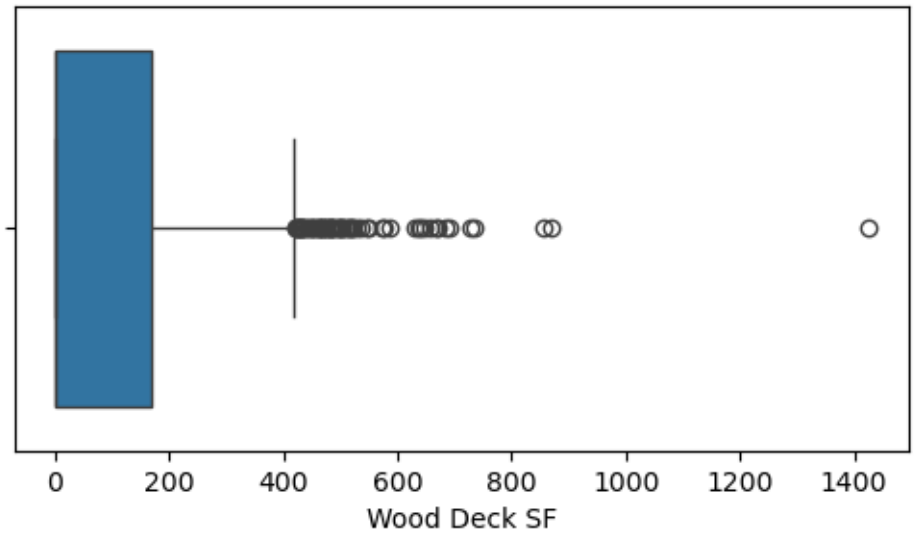
800
1000
1200
1400
Garage Area


0
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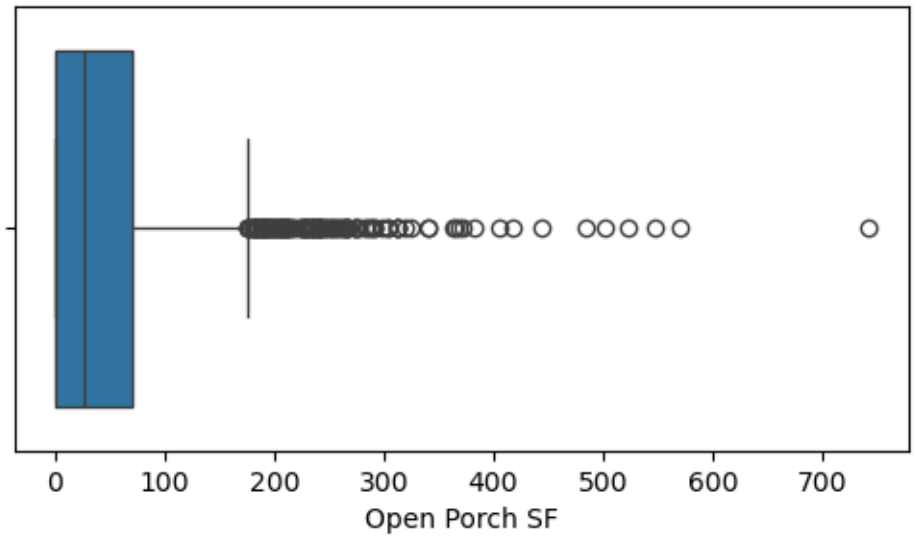
800
1000
1200
1400
Wood Deck SF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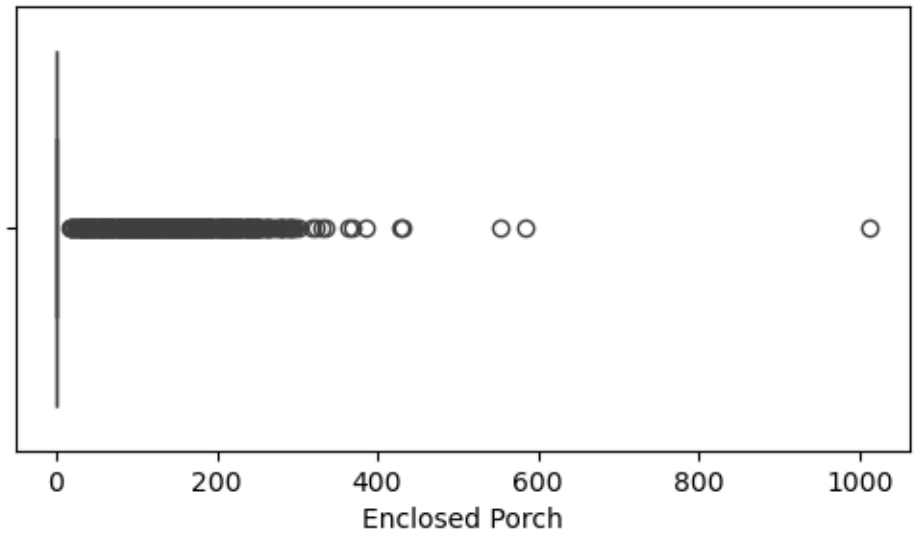
100
200
300
400
500
600
700
Open Porch SF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200
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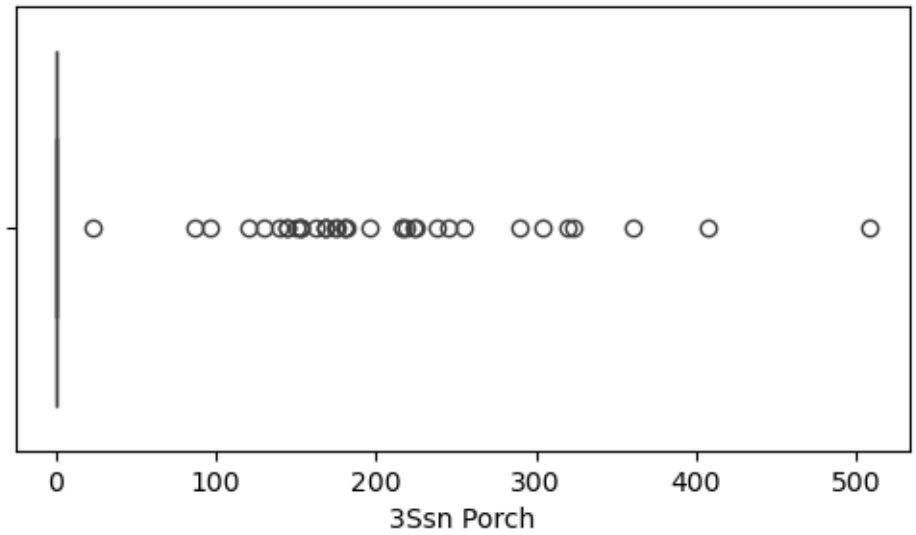
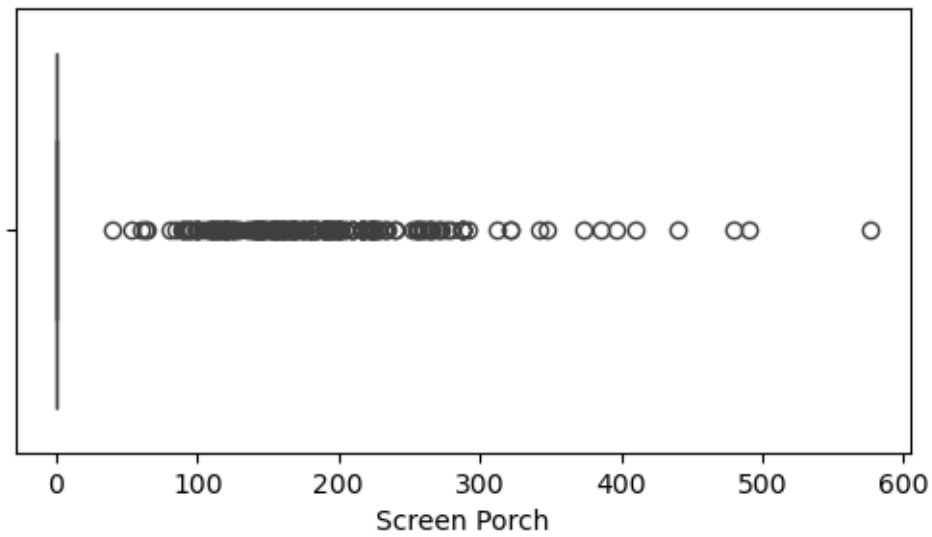
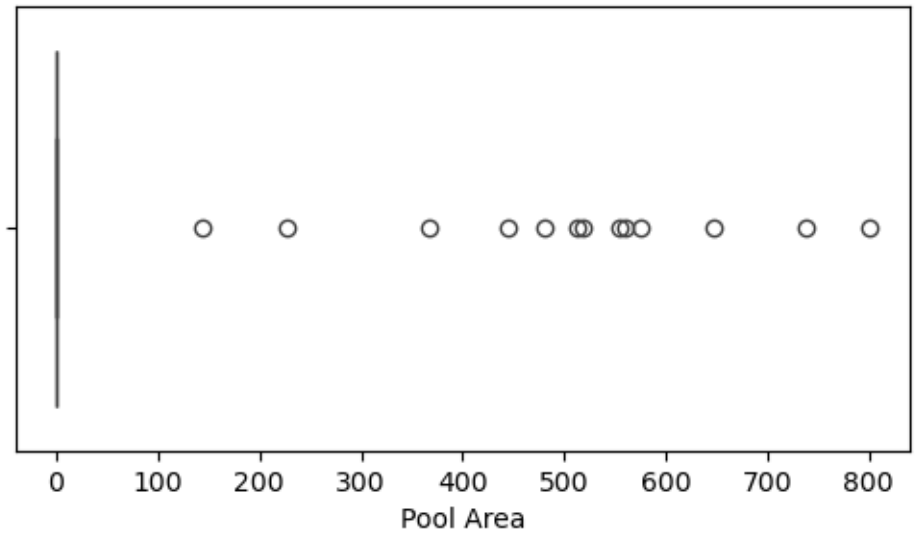
600
800
1000
Enclosed Porch


0
100
200
300
400
500
3Ssn Porch


0
100
200
300
400
500
600
Screen Porch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100
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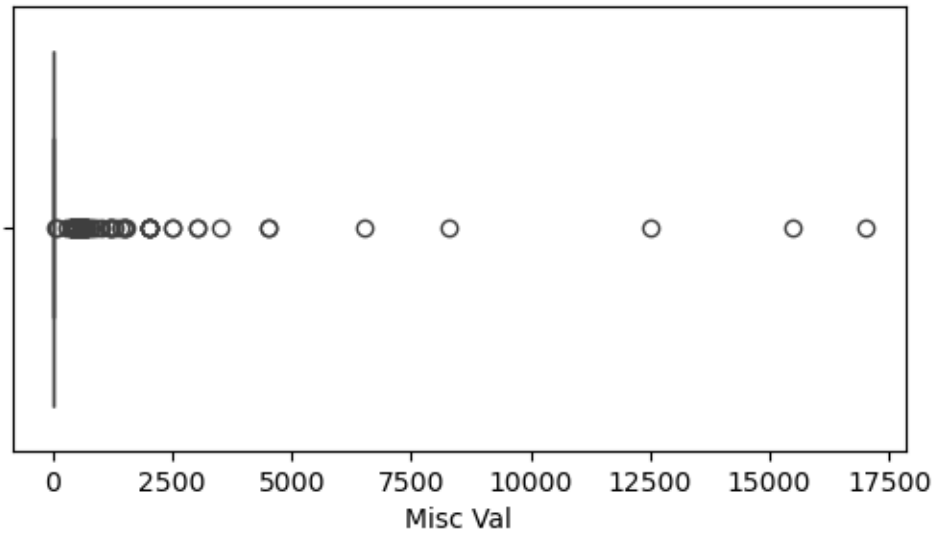
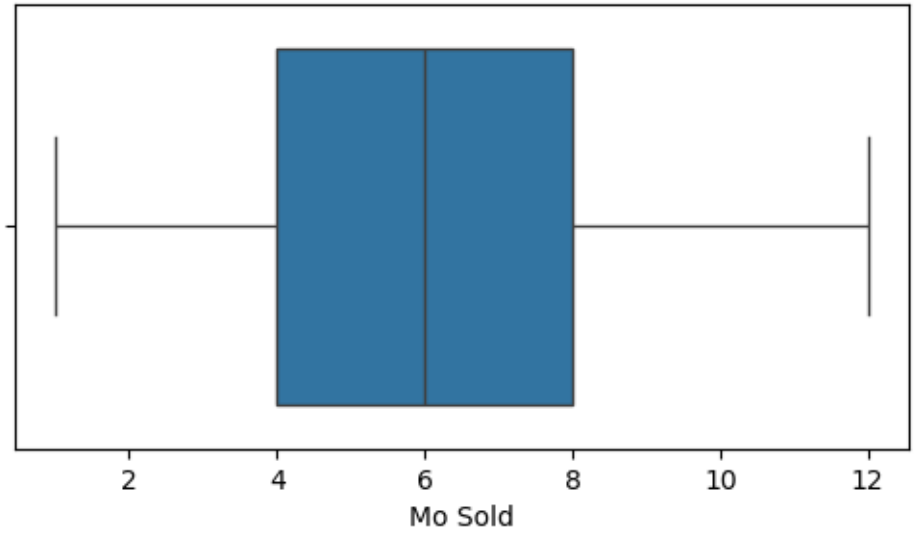
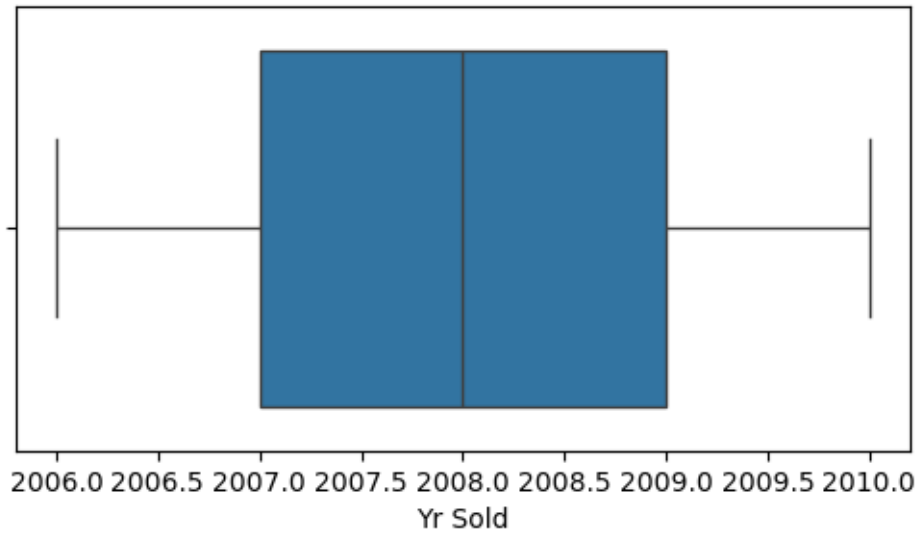
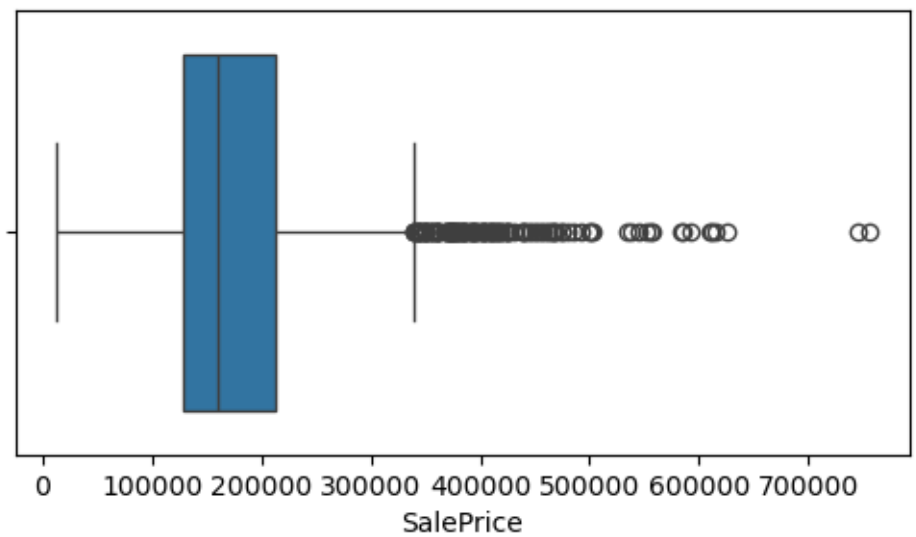
400
500
600
700
800
Pool Area


0
2500
5000
7500
10000
12500
15000
17500
Misc Val


2
4
6
8
10
12
Mo Sold



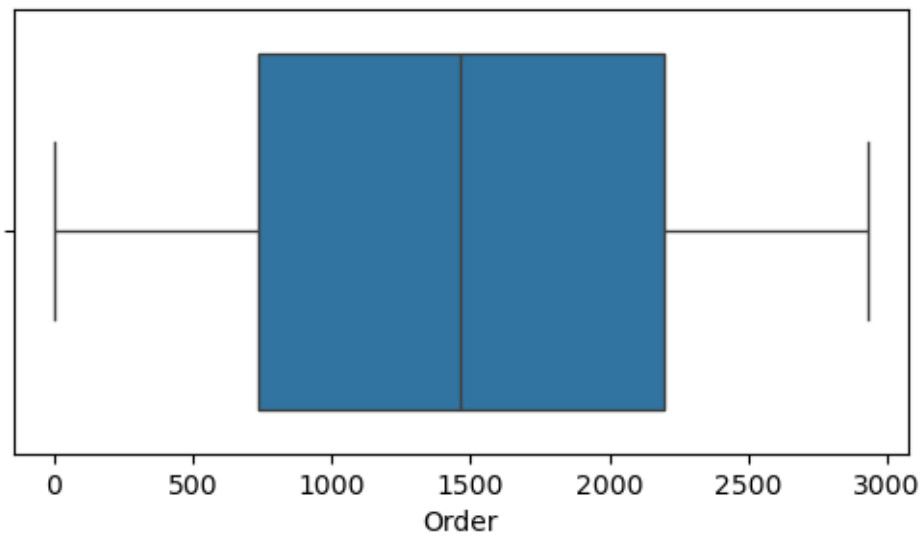


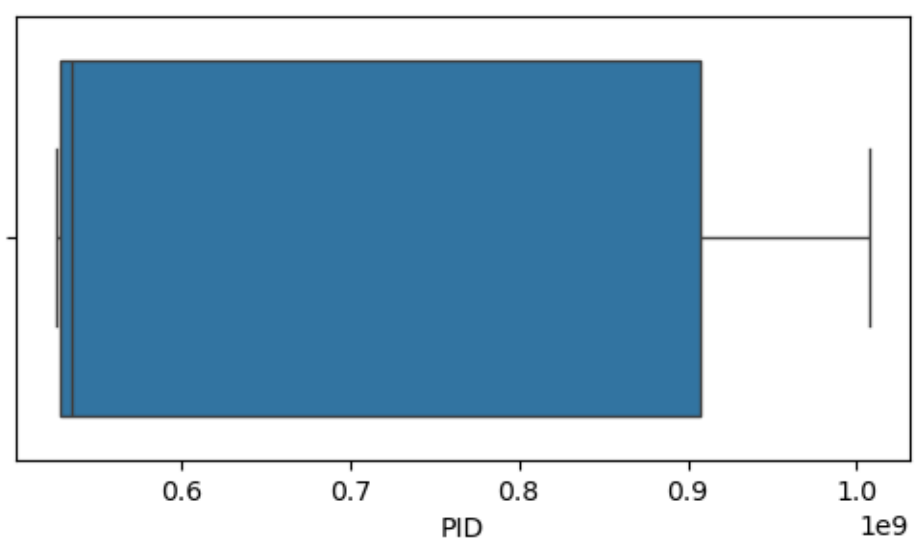
4.0.1 Replaced outliers with the lower and upper bound

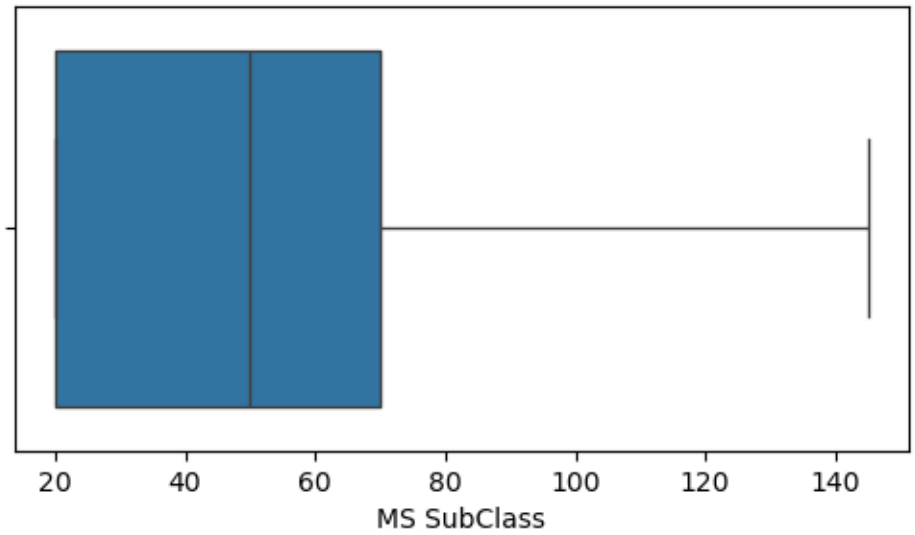
[54]:
```
# Replaced outliers with the lower and upper bound
num_cols = df.select_dtypes(include='number').columns
for i in num_cols:
    Q1 = df[i].quantile(0.25)
    Q3 = df[i].quantile(0.75)
    IQR = Q3-Q1
    lower = Q1-1.5*IQR
```

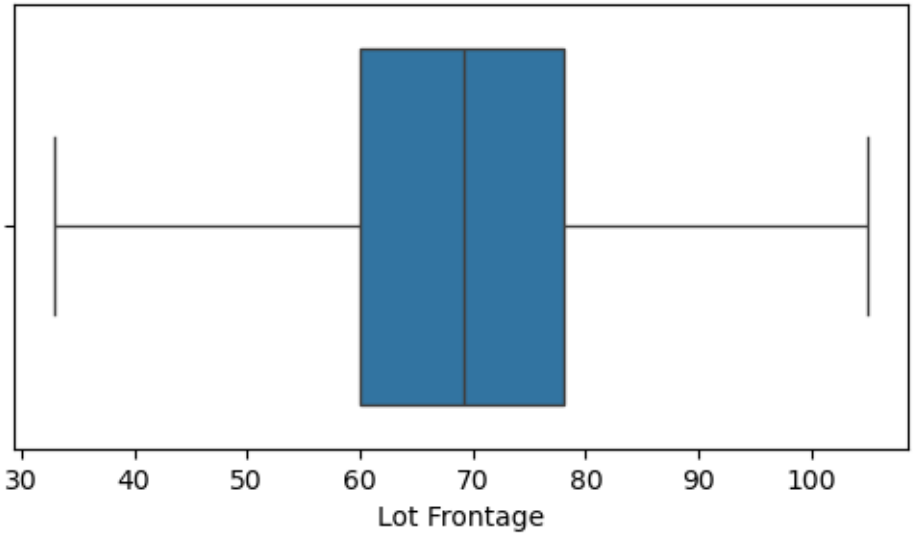
```
    upper = Q3+1.5*IQR
    df[i] = df[i].clip(lower,upper)
```

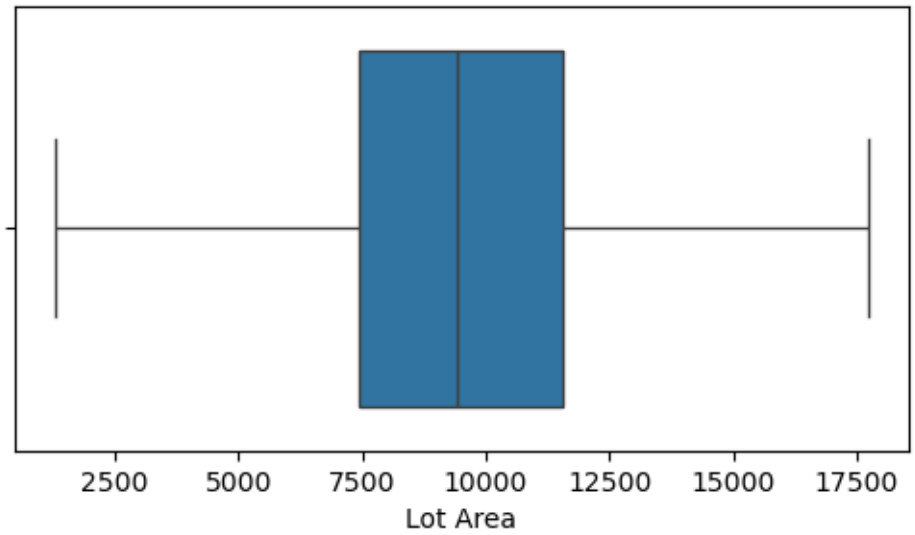
[55]:
```
for i in num_cols:
    plt.figure(figsize=(6,3))
    sns.boxplot(x=df[i])
    plt.show()
```

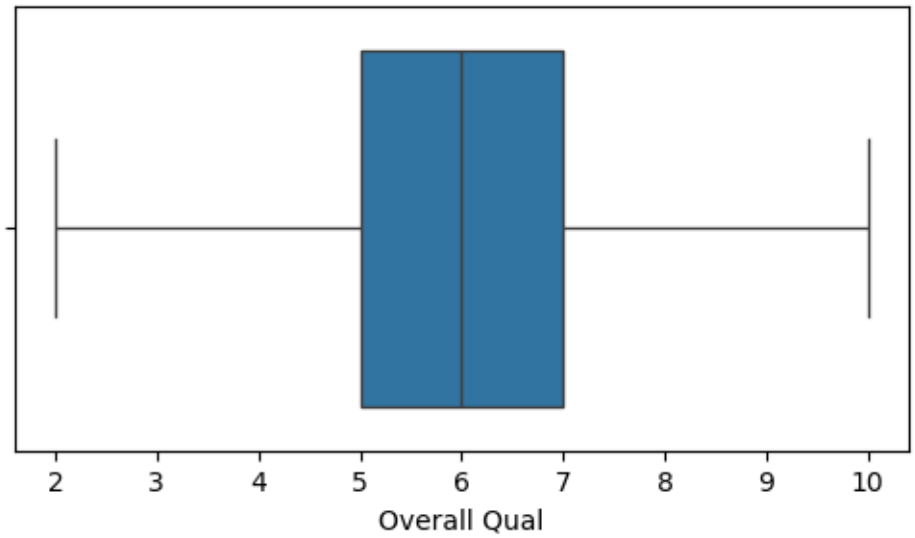


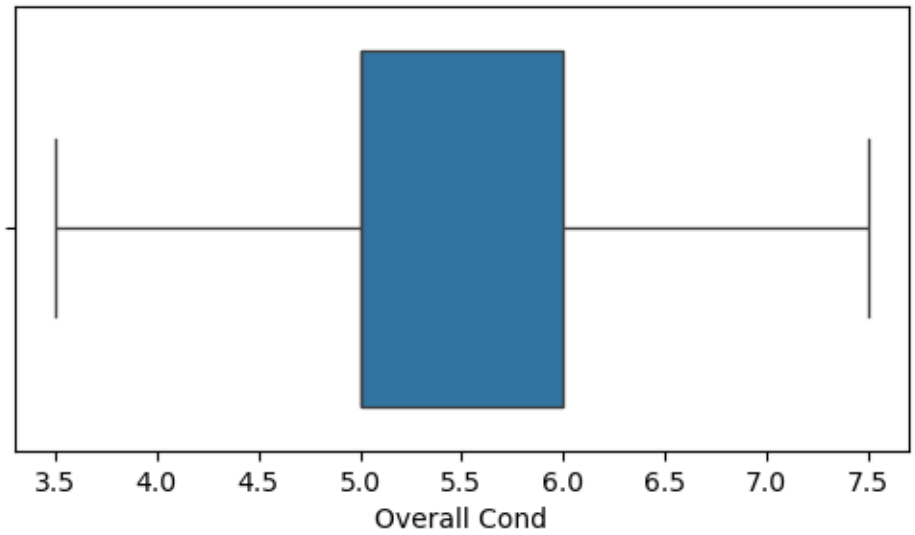


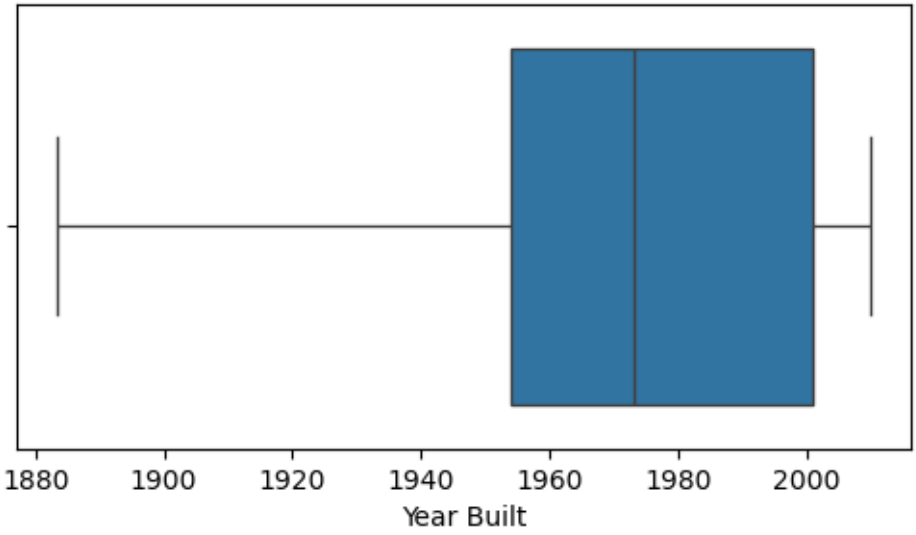

20
40
60
80
100
120
140
MS SubClass

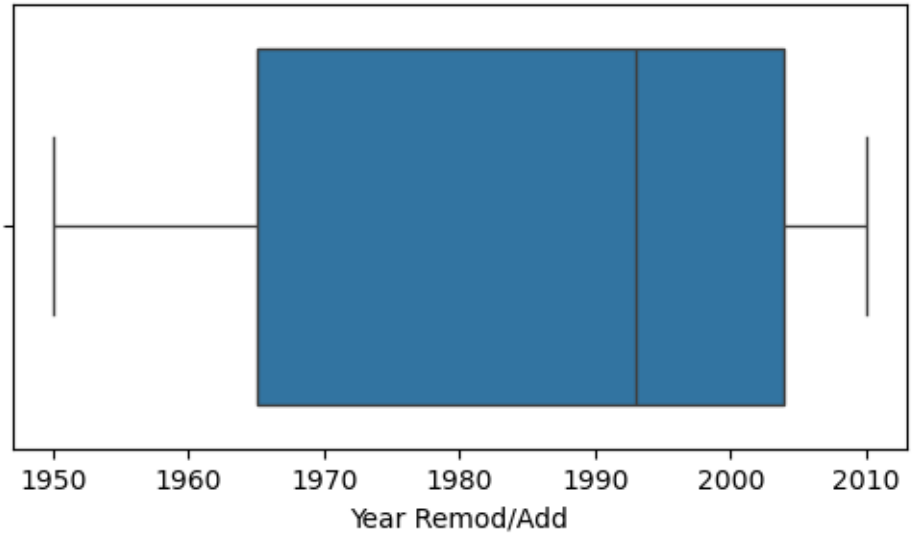

30
40
50
60
70
80
90
100
Lot Frontage

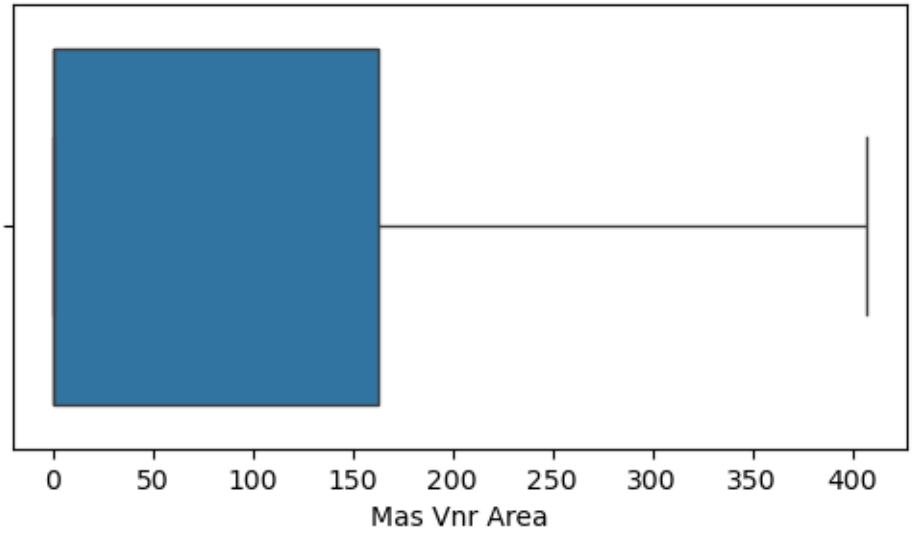

2500
5000
7500
10000
12500
15000
17500
Lot Area

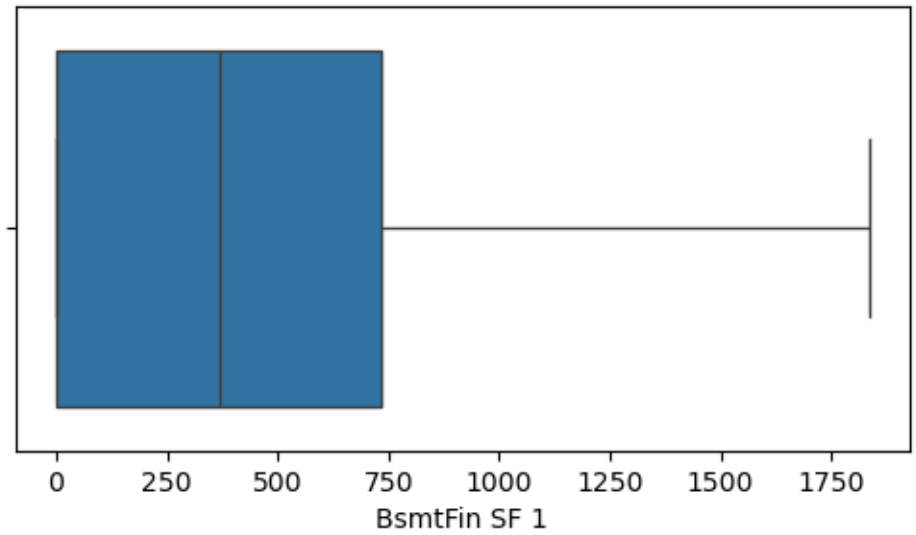

2
3
4
5
6
7
8
9
10
Overall Qual

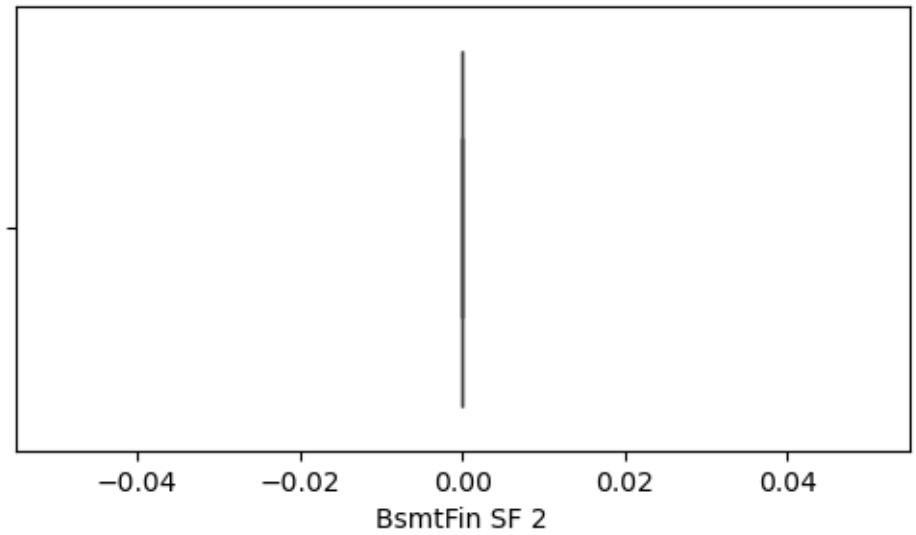

3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
Overall Cond

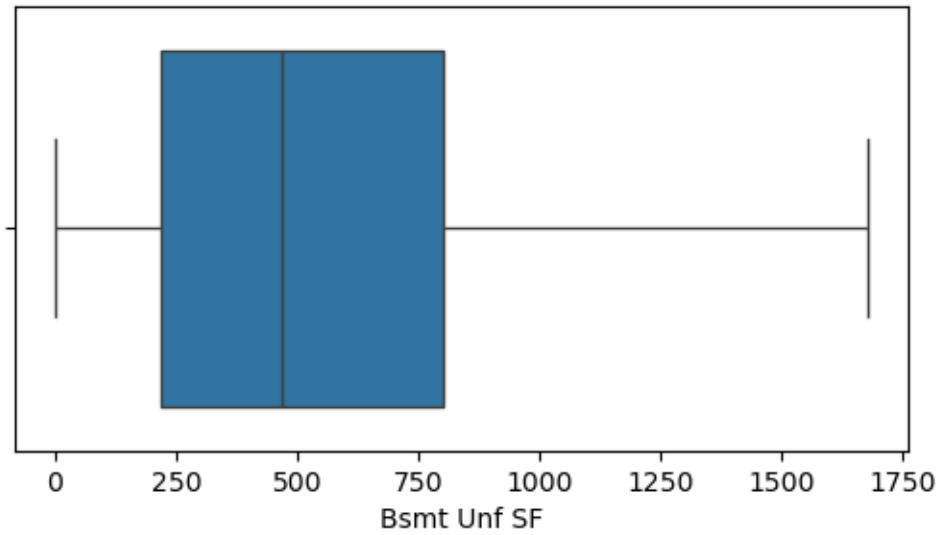

1880
1900
1920
1940
1960
1980
2000
Year Built

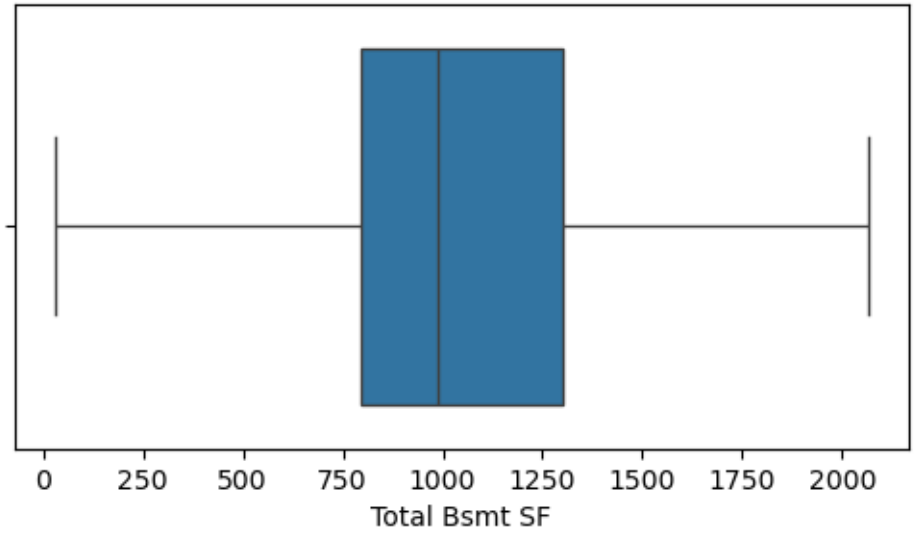

1950
1960
1970
1980
1990
2000
2010
Year Remod/Add

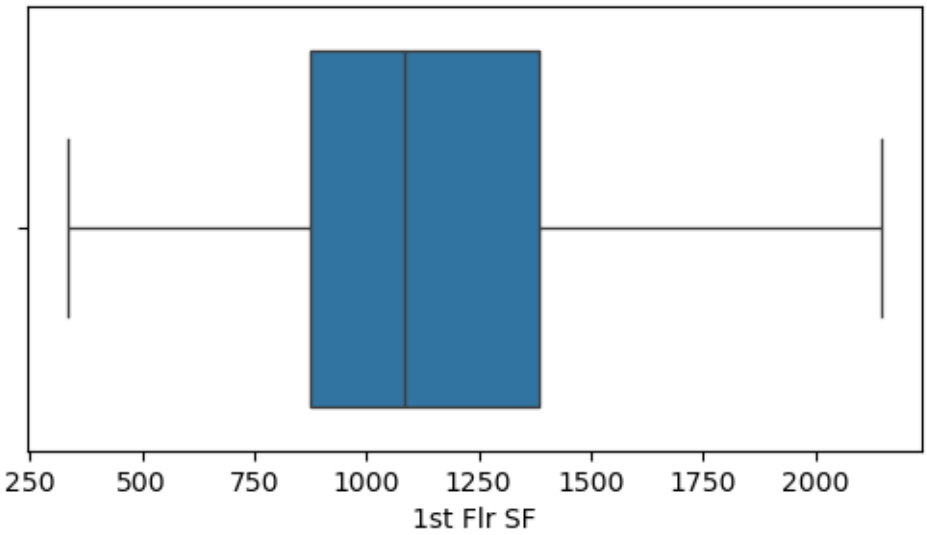

0
50
100
150
200
250
300
350
400
Mas Vnr Area

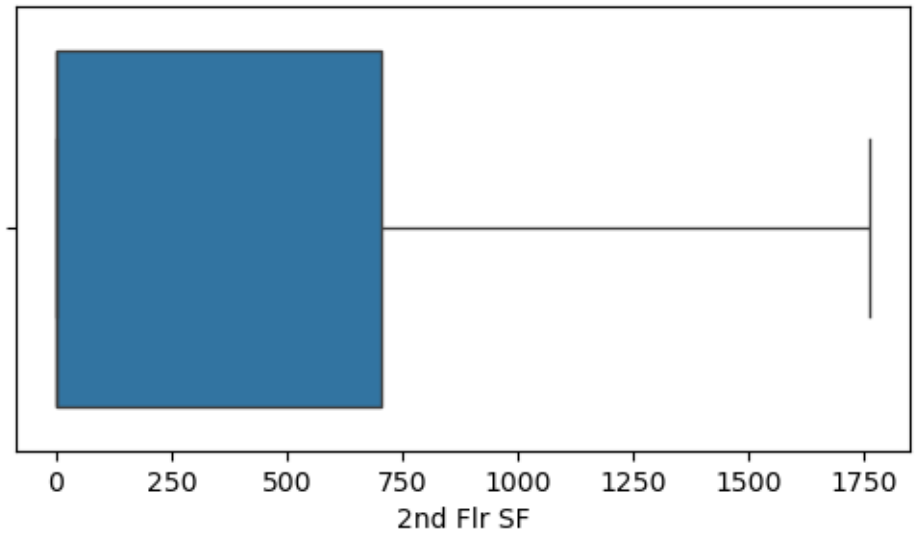

0
250
500
750
1000
1250
1500
1750
BsmtFin SF 1

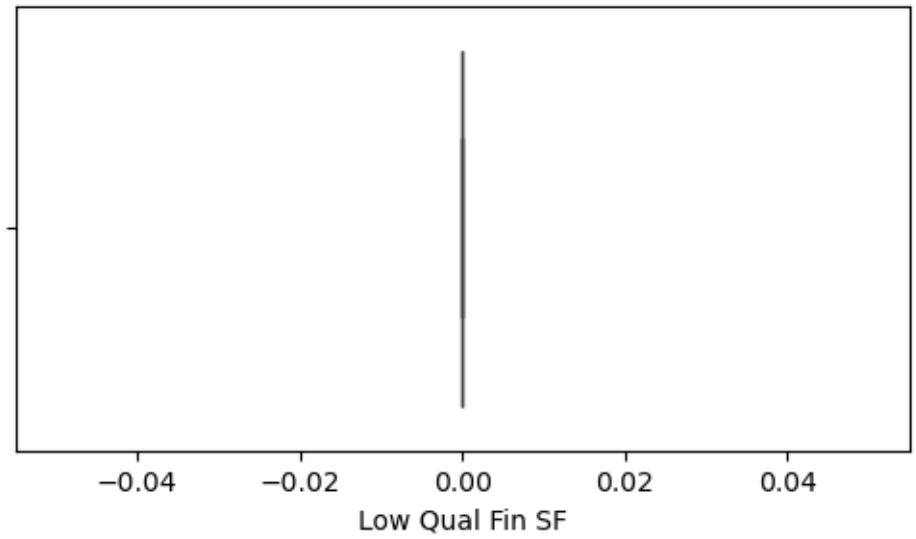

−0.04
−0.02
0.00
0.02
0.04
BsmtFin SF 2

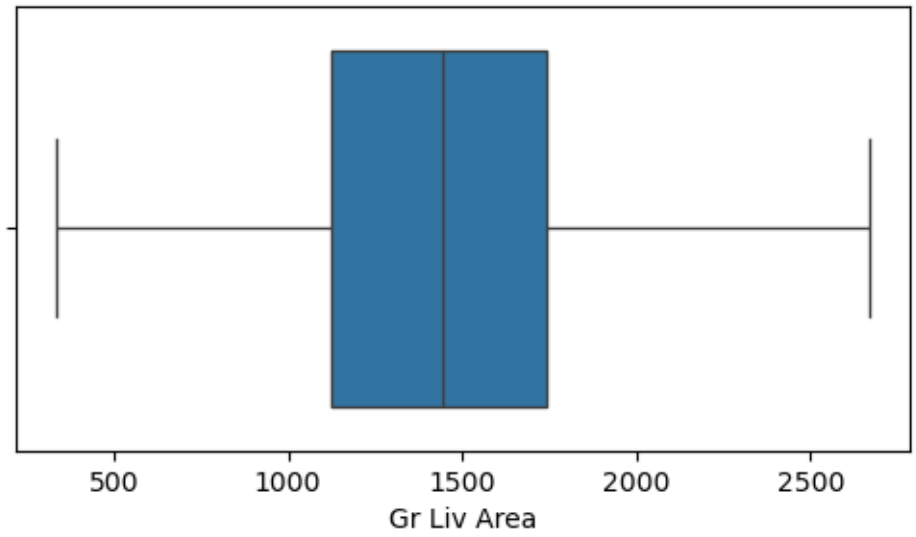

0
250
500
750
1000
1250
1500
1750
Bsmt Unf SF

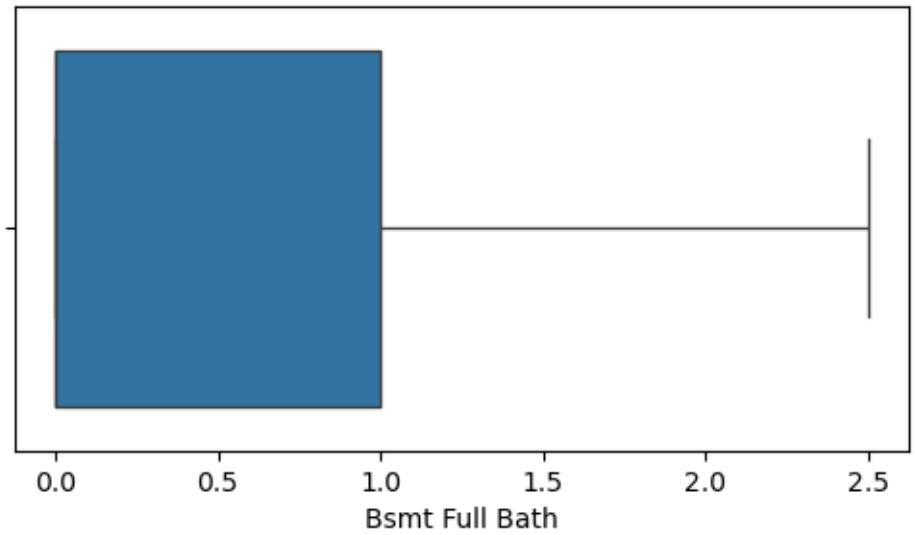

0
250
500
750
1000
1250
1500
1750
2000
Total Bsmt SF

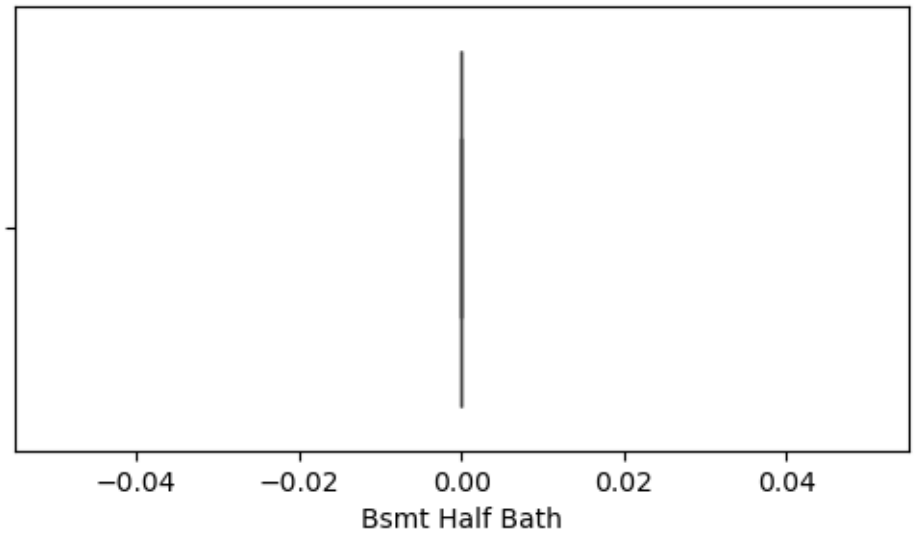

250
500
750
1000
1250
1500
1750
2000
1st Flr SF

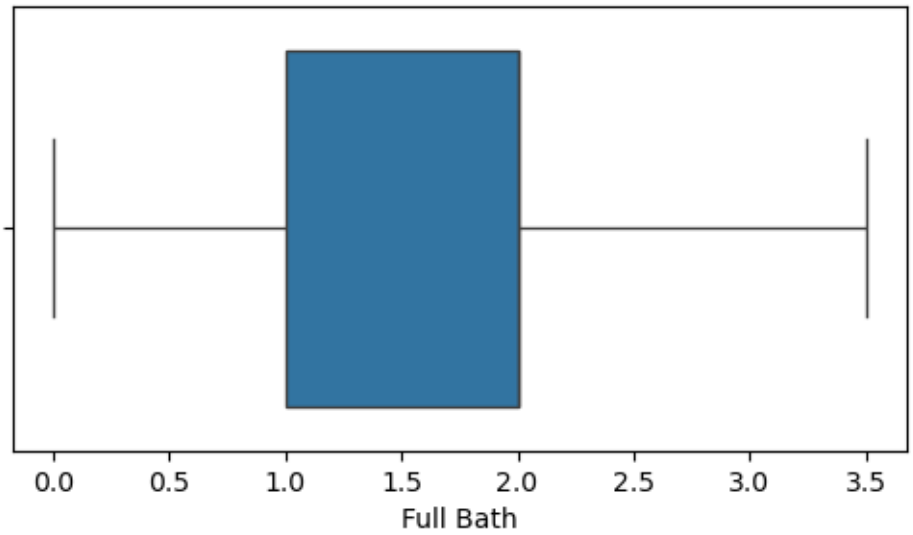

0
250
500
750
1000
1250
1500
1750
2nd Flr SF


−0.04
−0.02
0.00
0.02
0.04
Low Qual Fin SF


500
1000
1500
2000
2500
Gr Liv Area


0.0
0.5
1.0
1.5
2.0
2.5
Bsmt Full Bath


−0.04
−0.02
0.00
0.02
0.04
Bsmt Half Bath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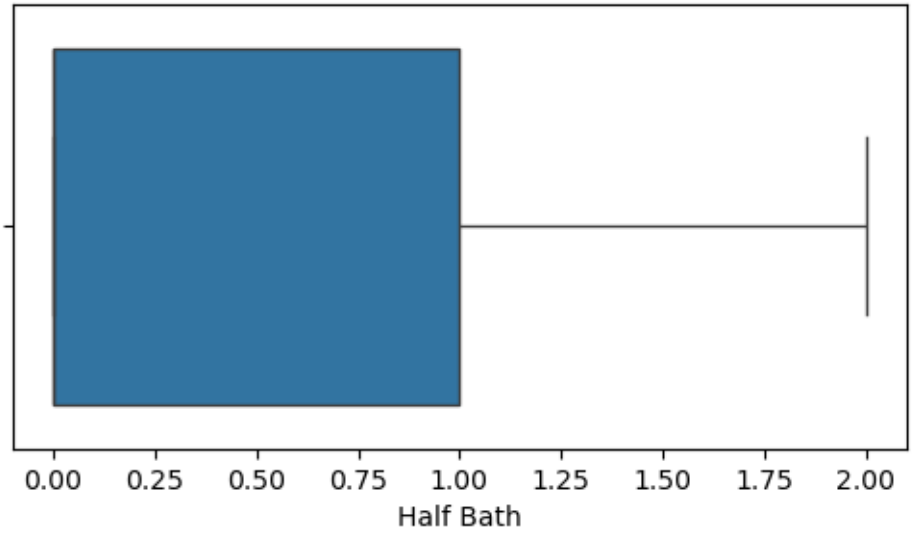
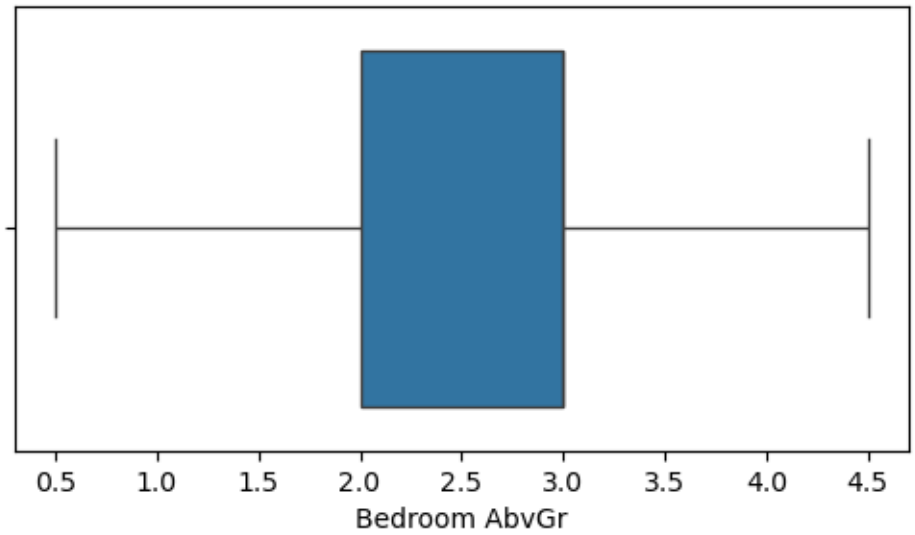
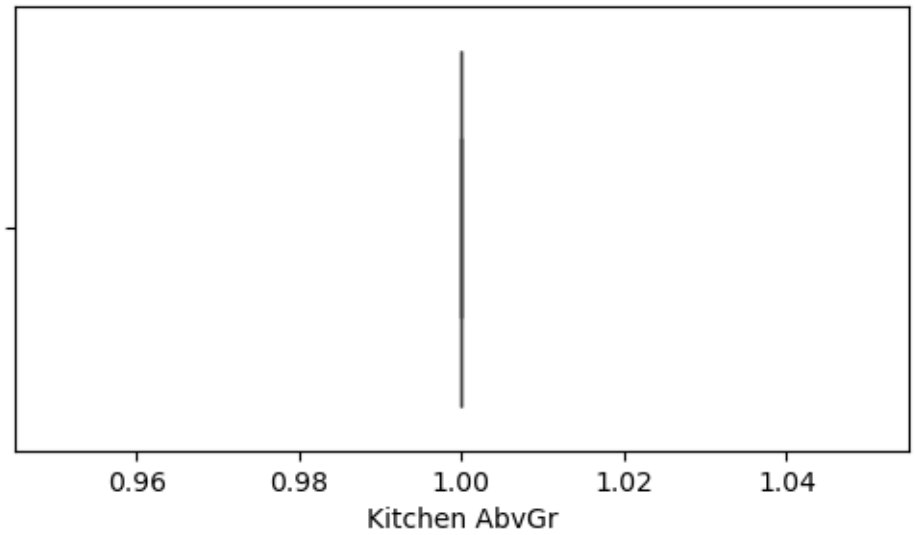
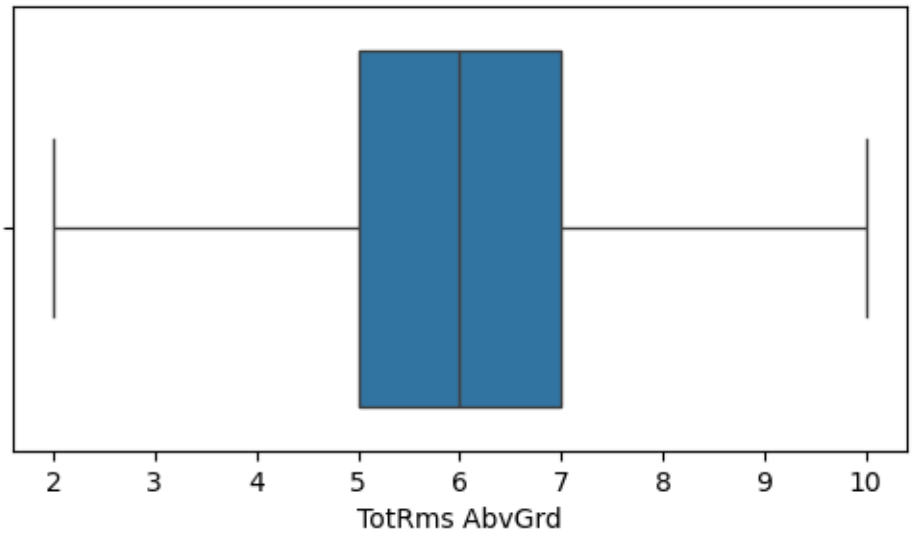
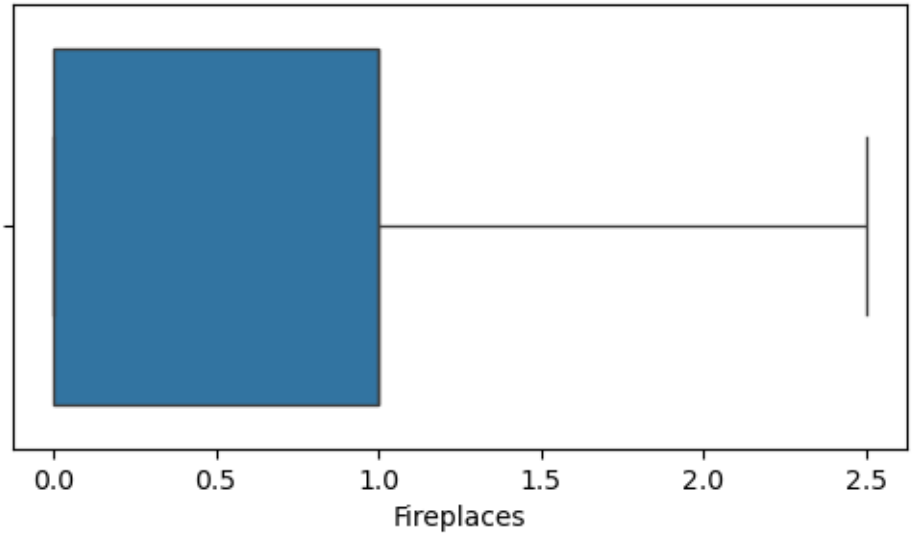
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Full Bath

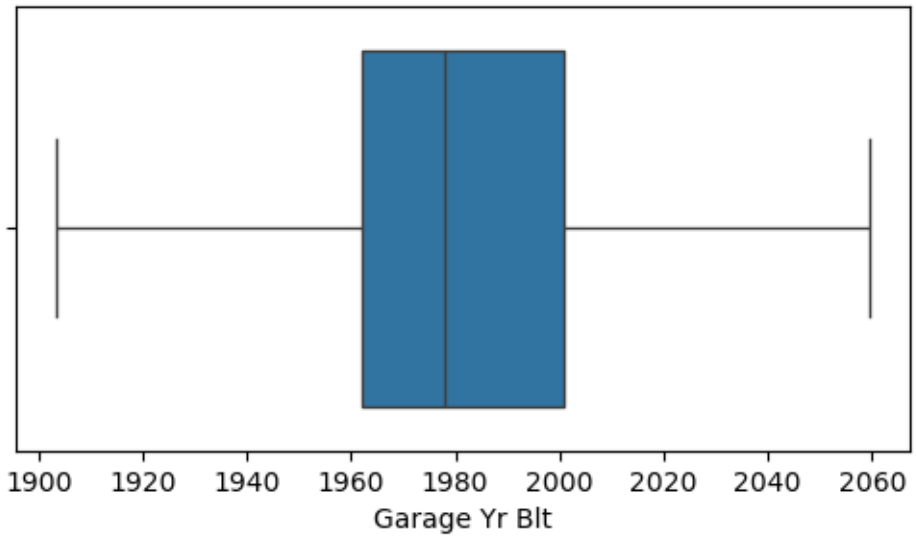

0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Half Bath

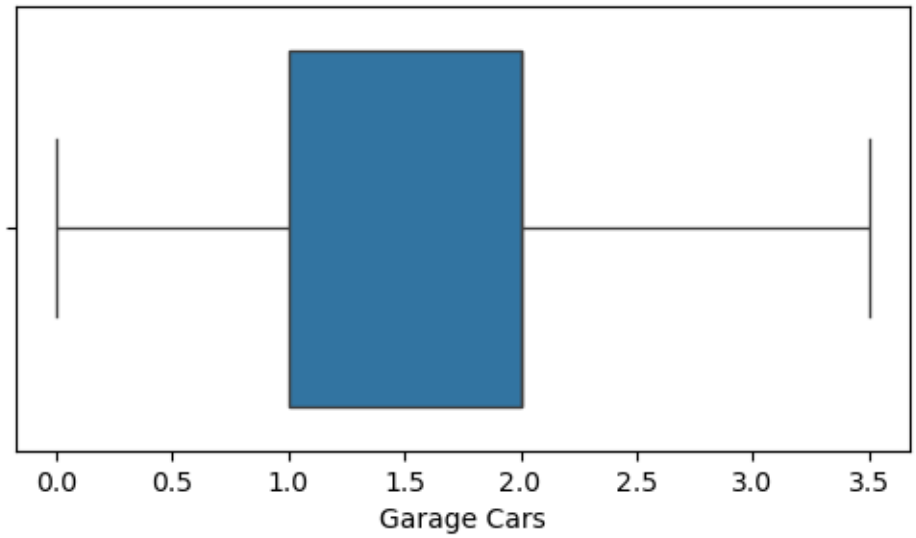

0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Bedroom AbvGr

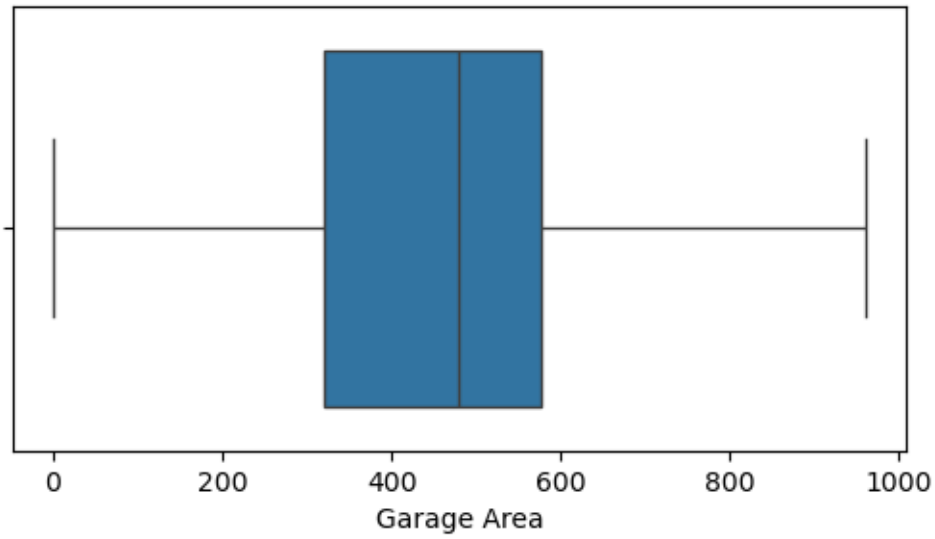

0.96
0.98
1.00
1.02
1.04
Kitchen AbvGr

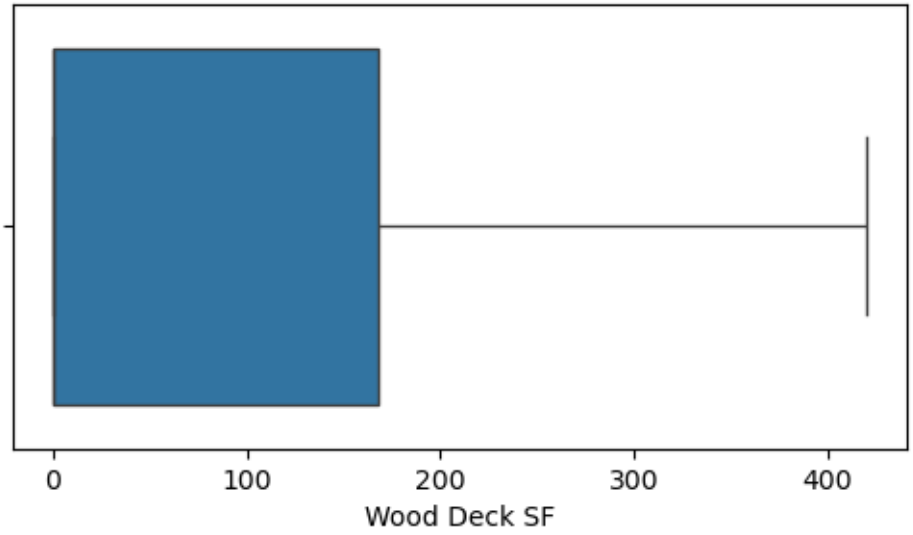

2
3
4
5
6
7
8
9
10
TotRms AbvGrd

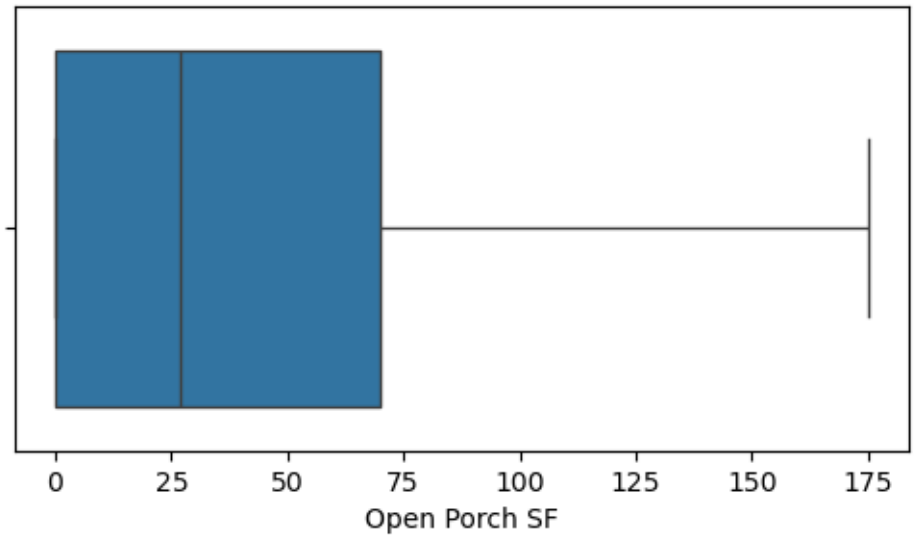

0.0
0.5
1.0
1.5
2.0
2.5
Fireplaces

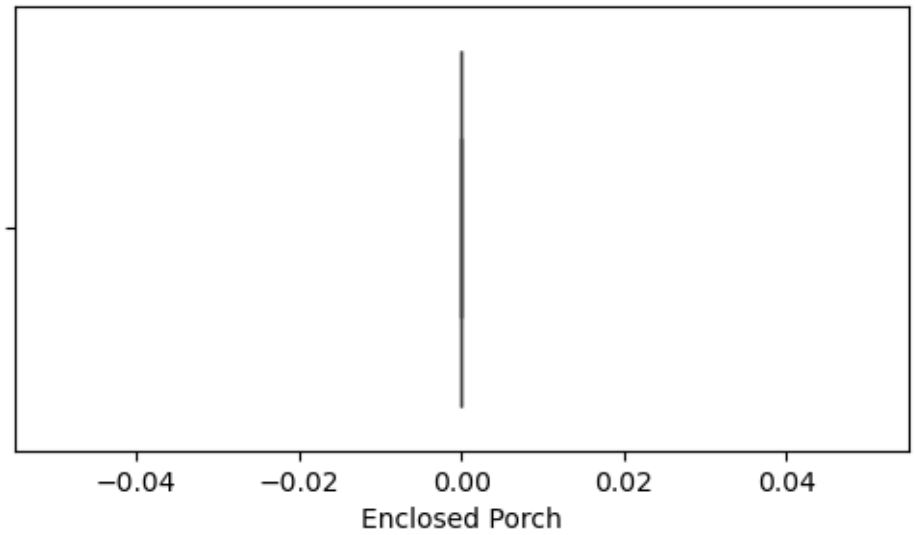

1900
1920
1940
1960
1980
2000
2020
2040
2060
Garage Yr Blt

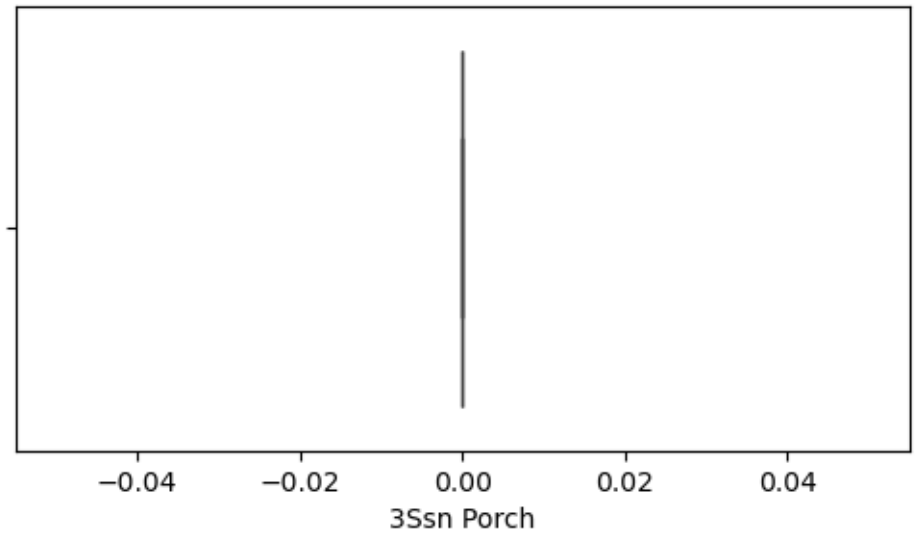

0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Garage Cars

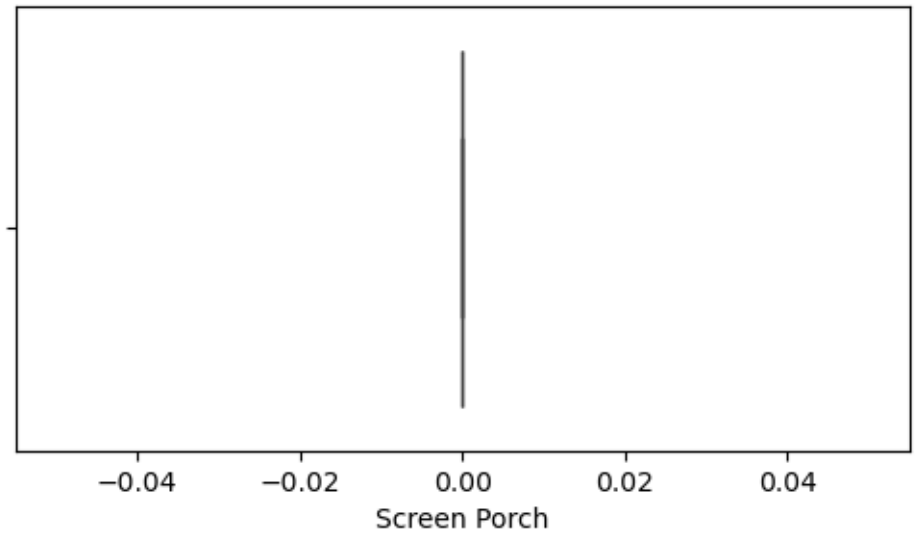

0
200
400
600
800
1000
Garage Area

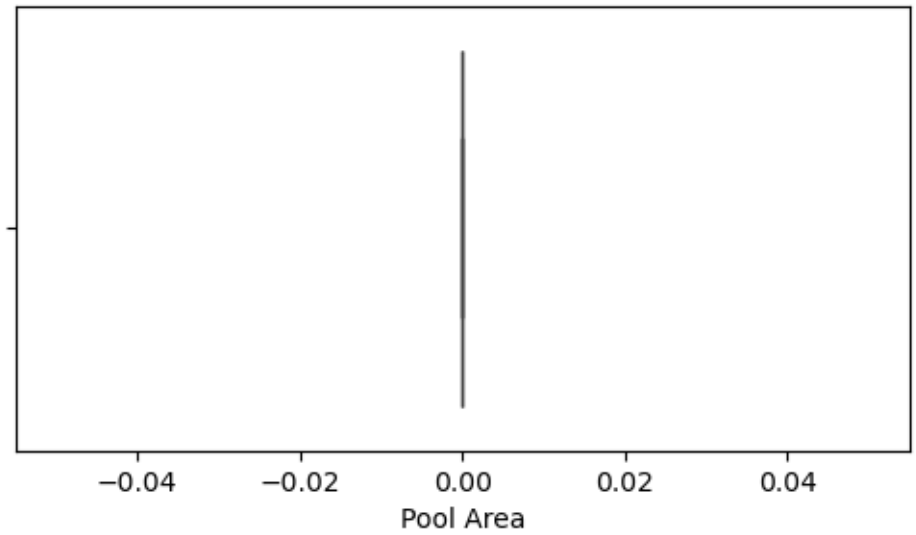

0
100
200
300
400
Wood Deck SF

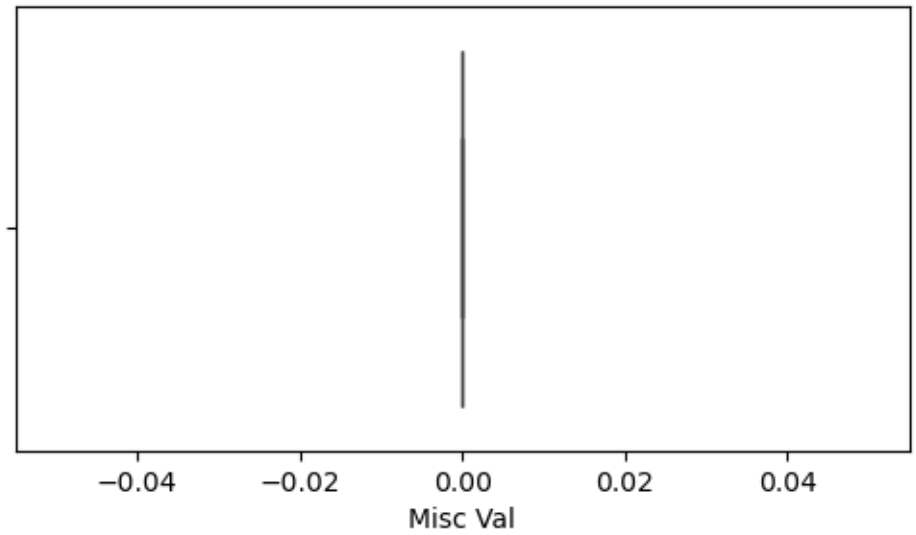

0
25
50
75
100
125
150
175
Open Porch SF

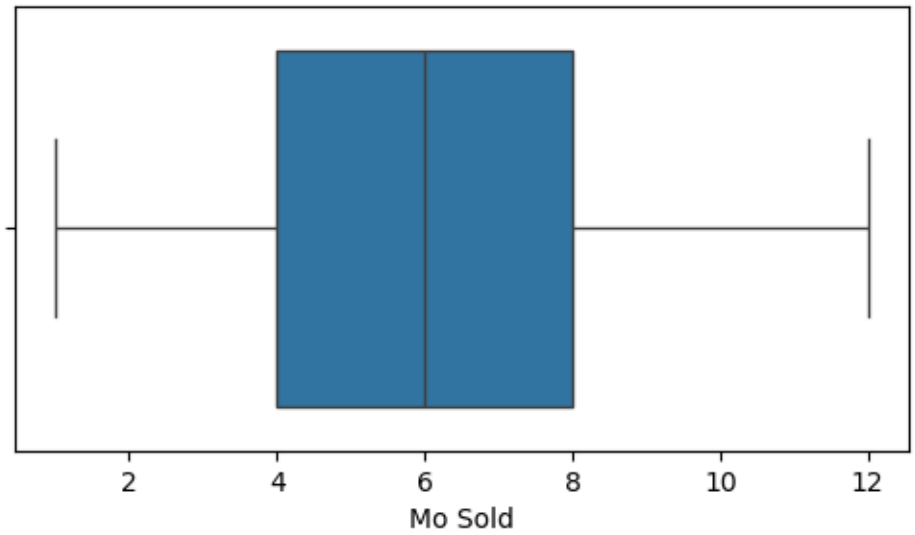

−0.04
−0.02
0.00
0.02
0.04
Enclosed Porch

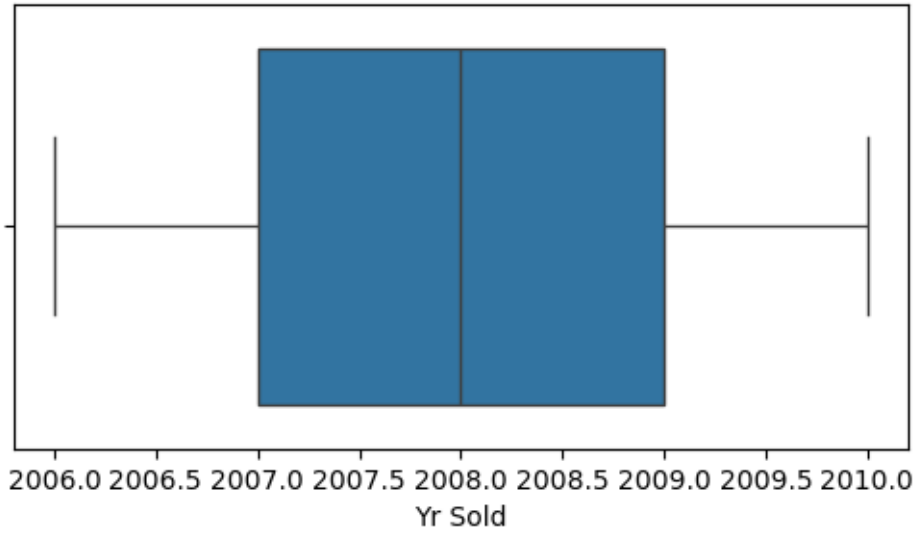

−0.04
−0.02
0.00
0.02
0.04
3Ssn Porch

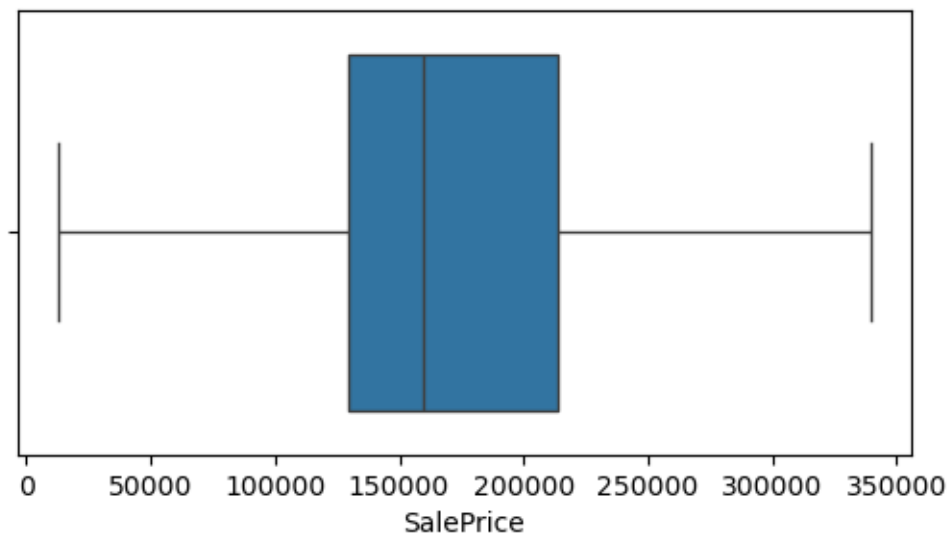

−0.04
−0.02
0.00
0.02
0.04
Screen Porch


−0.04
−0.02
0.00
0.02
0.04
Pool Area


−0.04
−0.02
0.00
0.02
0.04
Misc Val

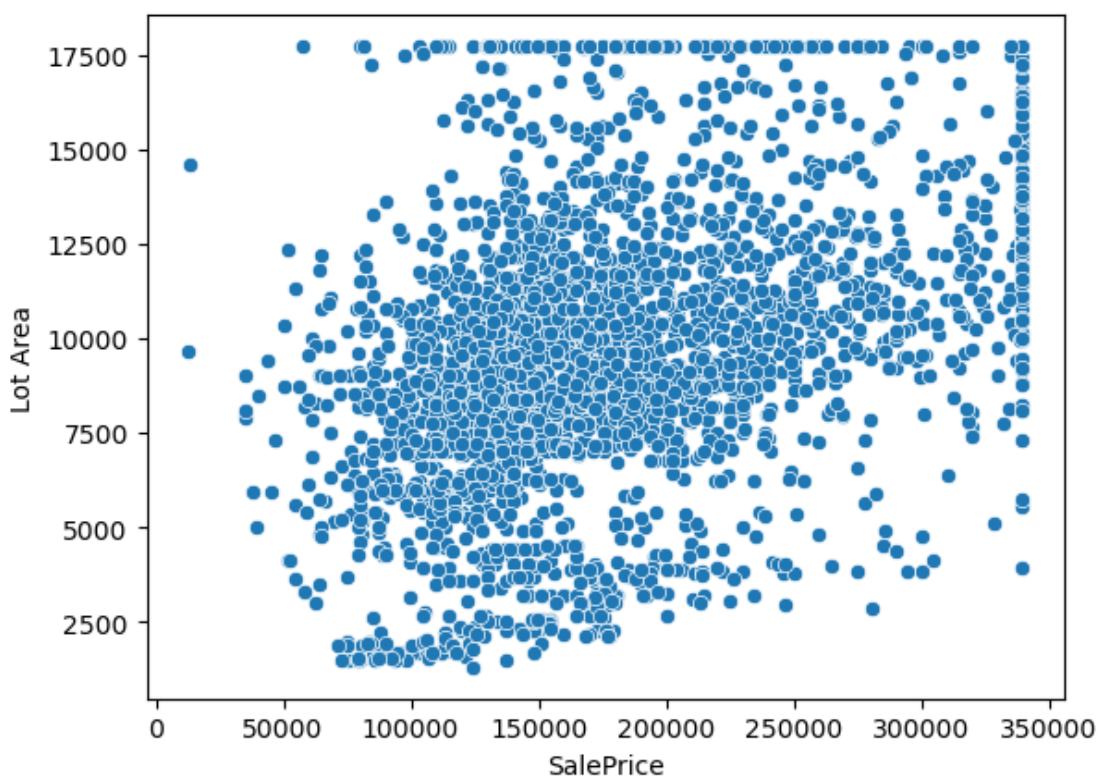

2
4
6
8
10
12
Mo Sold


2006.0 2006.5 2007.0 2007.5 2008.0 2008.5 2009.0 2009.5 2010.0
Yr Sold



[56]:
```
# Scatter plot for area vs price
sns.scatterplot(x='SalePrice',y='Lot Area',data=df)
```

[56]: <Axes: xlabel='SalePrice', ylabel='Lot Area'>



[46]:
```
df.drop(columns=['Order','PID'],inplace=True)
```

Encoding categorical features(Location,type)

[53]:
```
str_col = df.select_dtypes(include='object').columns
str_col

# Encoding categorical features(Location,type)
from sklearn.preprocessing import LabelEncoder
encoder = LabelEncoder()
for i in str_col:
    df[i] = encoder.fit_transform(df[i])
```

[55]:
```
df
```

[55]:
```
      MS SubClass  MS Zoning  Lot Frontage  Lot Area  Street  Lot Shape  \
0              20          5     141.00000     31770       1          0
1              20          4      80.00000     11622       1          3
2              20          5      81.00000     14267       1          0
3              20          5      93.00000     11160       1          3
4              60          5      74.00000     13830       1          0
…           …         …            …       …      …         …
2925           80          5      37.00000      7937       1          0
2926           20          5      69.22459      8885       1          0
2927           85          5      62.00000     10441       1          3
2928           20          5      77.00000     10010       1          3
2929           60          5      74.00000      9627       1          3

      Land Contour  Utilities  Lot Config  Land Slope  …  Pool Area  \
0                3          0           0           0  …          0
1                3          0           4           0  …          0
2                3          0           0           0  …          0
3                3          0           0           0  …          0
4                3          0           4           0  …          0
…             …         …          …          …  …         …
2925             3          0           1           0  …          0
2926             2          0           4           1  …          0
2927             3          0           4           0  …          0
2928             3          0           4           1  …          0
2929             3          0           4           1  …          0

      Pool QC  Fence  Misc Feature  Misc Val  Mo Sold  Yr Sold  Sale Type  \
0           0      2             3         0        5     2010          9
1           0      2             3         0        6     2010          9
2           0      2             1     12500        6     2010          9
3           0      2             3         0        4     2010          9
```

```
4            0       2              3          0         3      2010           9
…        …     …            …        …       …       …         …
2925         0       0              3          0         3      2006           9
2926         0       2              3          0         6      2006           9
2927         0       2              3        700         7      2006           9
2928         0       2              3          0         4      2006           9
2929         0       2              3          0        11      2006           9

      Sale Condition  SalePrice
0                  4     215000
1                  4     105000
2                  4     172000
3                  4     244000
4                  4     189900
…               …         …
2925               4     142500
2926               4     131000
2927               4     132000
2928               4     170000
2929               4     188000

[2930 rows x 77 columns]
```

[57]:
```
X = df.iloc[:,0:72].values
y=df.iloc[:,-1].values
```

[59]:
```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size = 0.25)
```

4.0.2 Feature Scaling

[62]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

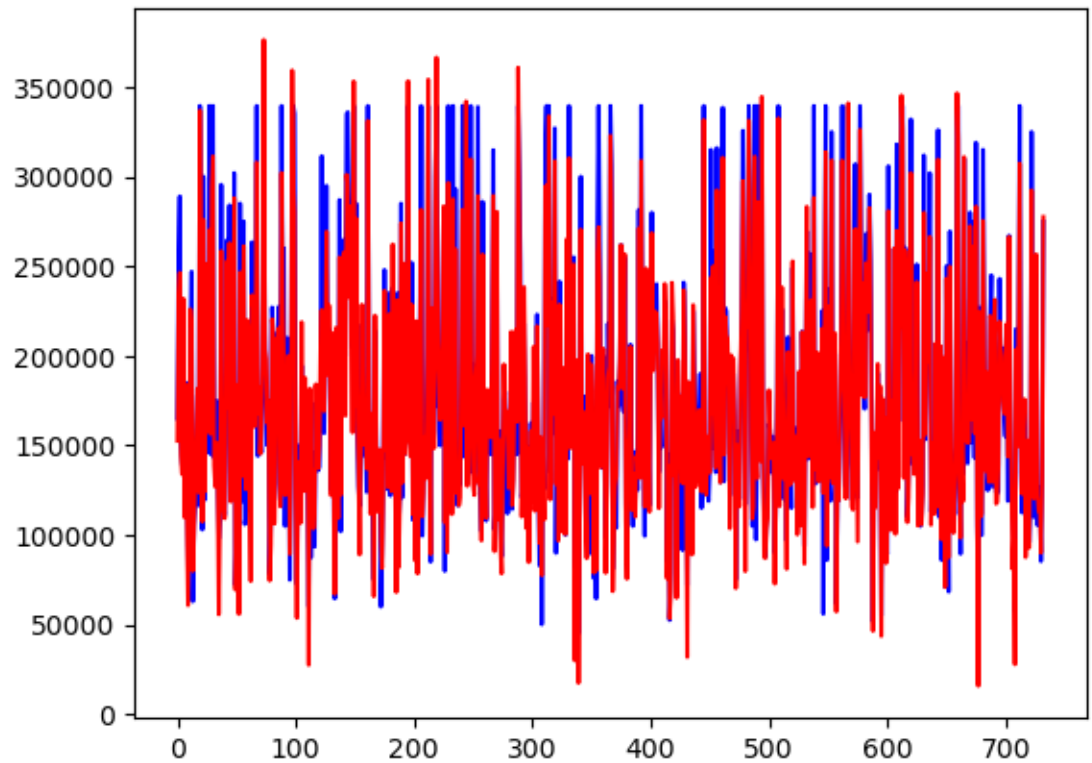
4.0.3 LinearRegression

[65]:
```
from sklearn.linear_model import LinearRegression
reg = LinearRegression()
reg.fit(X_train,y_train)
```

[65]:
```
LinearRegression()
```

[67]:
```
y_pred = reg.predict(X_test)
```

[82]:
```
plt.plot(y_test,color='blue',label='Test')
plt.plot(y_pred,color='red',label='Prediction')
plt.show()
```



[84]:
```
import numpy as np
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
print('R2 Score: ',r2_score(y_test,y_pred))
print('MAE: ',mean_absolute_error(y_test,y_pred))

print('RMSE: ',np.sqrt(mean_squared_error(y_test,y_pred)))
```

```
R2 Score:  0.9213771528307929
MAE:  15147.556376227036
RMSE:  19944.28429273191
```

[86]:
```
data = [[20,5,81.0,14267.0,1,0,3,0,0,0,15,2,2,0,2,6,6.0,1958.
 ↪0,1958,5,1,14,15,108.0,3,10,1,4,4,3,0,923.0,5,0.0,406.0,1329.0,1,4,1,4,1329.
 ↪0,0.0,0,1329.0,0.0,0.0,1.0,1,3.0,1,2,6,7,0.0,1,1958.0,2,1.0,312.
 ↪0,4,4,2,393,36,0,0,0,0,0,6,2010,9]]
new_df = pd.DataFrame(data)
new_val = scaler.transform(new_df)
predicted_value = reg.predict(new_val)
```

```
print(predicted_value.round())
```

```
[169142.]
```

5 Decision Tree

[93]:
```
from sklearn.tree import DecisionTreeClassifier
DT=DecisionTreeClassifier()
DT.fit(X_train,y_train)
```

[93]:
```
DecisionTreeClassifier()
```

[95]:
```
y_pred = DT.predict(X_test)
```

[97]:
```
from sklearn.metrics import accuracy_score, confusion_matrix
accuracy = accuracy_score(y_test, y_pred)
confusion_matrix  = confusion_matrix(y_test, y_pred)
print(accuracy)
print(confusion_matrix)
```

```
0.040927694406548434
[[ 0  0  0 …  0  0  0]
 [ 0  0  0 …  0  0  0]
 [ 0  0  0 …  0  0  0]
 …
 [ 0  0  0 …  0  0  1]
 [ 0  0  0 …  0  0  0]
 [ 0  0  0 …  0  1 23]]
```

[99]:
```
from sklearn.ensemble import RandomForestClassifier

# initiate the classifier
model_rf = RandomForestClassifier()

# Passing the data to classifier
model_rf.fit(X_train,y_train)
```

[99]:
```
RandomForestClassifier()
```

[100]:
```
y_pred_rf = model_rf.predict(X_test)
```

[106]:
```
from sklearn.metrics import accuracy_score
print(accuracy_score(y_test,y_pred_rf))
```

```
0.06002728512960437
```

[ ]:
```

```

[ ]:

```

```